SPFormer: Self-Pooling Transformer for Few-Shot Hyperspectral Image Classification

Ziyu Li, Zhaohui Xue, *Member, IEEE*, Qi Xu, Ling Zhang, Tianzhi Zhu, and Mengxue Zhang

***Abstract*— Transformer has shown great potential in extracting global features, and it can achieve better classification performance with a large number of training samples compared with other deep learning (DL) models. However, most of the existing Transformer-based models for hyperspectral image (HSI) classification simply use the multihead self-attention and channel multilayer perceptron (MLP) modules that contain many parameters to learn, resulting in poor performance in a few-shot learning scenario. To overcome the above issue, a lightweight self-pooling Transformer (SPFormer) is proposed for few-shot HSI classification. First, a one-layer autoencoder based on self-supervised learning is built to reduce the dimensionality of HSI. Second, two parameter-free modules, channel shuffle for multihead self-pooling with sparse mapping (CSSM-MHSP) and central token mixer (CTM), are proposed for mapping spectral features to higher dimensions and promoting information interaction between pixels, respectively. Third, a lightweight channel embedding is designed to extract deep spectral features. Finally, a fully connected layer is used for classification. The classification performance of the proposed method is evaluated on four benchmark datasets, showing its superiority in classification accuracy, generalization performance, and model complexity compared to existing state-of-the-art methods with limited training samples.**

***Index Terms*— Autoencoder, channel shuffle (CS), few-shot learning (FSL), hyperspectral image (HSI) classification, self-pooling, Transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) captures rich spectral information of land cover across hundreds of continuous channels. With extremely high spectral resolution, HSI can distinguish different ground objects by their spectral differences [1]. Therefore, HSI classification is widely used in environmental monitoring, national defense, precision agriculture, mineral exploration [2], [3], [4], and so on.

In the past decade, classical machine learning methods, including support vector machine (SVM) [5], random forest (RF) [6], representation learning [7], and so on, have greatly promoted the development of HSI classification. Nevertheless, the majority of machine learning-based methods mentioned above focus on low-level features, such as spectral information or handcrafted features.

Deep learning (DL) has been extensively used in image recognition because it can automatically extract deep semantic features. Many classic DL frameworks have been explored for HSI classification, such as stacked autoencoder (SAE) [8], deep belief network (DBN) [9], convolutional neural network (CNN) [10], recurrent neural network (RNN) [11], long short-term memory (LSTM) [12], and generative adversarial network (GAN) [13]. Among them, CNN is the most widely used DL framework. According to different convolution kernel operation rules, CNN can be divided into 1-D CNN [14], 2-D CNN [15], and 3-D CNN [16], [17], [18], [19], [20]. Although CNN-based DL frameworks have achieved great performance, such methods cannot effectively mine the most important features and ignore irrelevant information about the image.

Attention mechanism, which can enhance valuable features and weaken the weight of noisy information [21], has been widely studied to enhance the classification performance of DL models. Generally, the attention mechanism can be directly embedded into DL frameworks as a plug-and-play module. Currently, most of the attention-based HSI classification methods focus on spectral attention [22], spatial attention [23], spectral–spatial attention [24], [25], [26], self-attention [27], and cross-attention [28].

Recently, an advanced attention-based DL framework, Transformer [29], emerged as a completely different approach from CNN. Transformer has a stronger ability to model long-distance dependencies, particularly demonstrated by the Vision Transformer (ViT) [30] and its variants outperforming CNN-based models in image recognition. Existing studies based on Transformer have exhibited great performance for HSI classification. He et al. [31] combined CNN and Transformer earlier, proposing spatial–spectral Transformer (SST) for HSI classification. Later, Hong et al. [32] examined the HSI classification process from the perspective of sequence data for the first time and proposed a novel model architecture called SpectralFormer (SF), which can extract local spectral features from adjacent spectral bands and uses a cross-layer skip connection to avoid information loss during training.

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To extract spectral–spatial features and high-level semantic features of HSI, Sun et al. [33] combined the convolution layer and Transformer structure to propose a spectral–spatial feature tokenization Transformer (SSFTT). Subsequently, in order to address the issue of overdispersion in the output features of Transformer, Mei et al. [34] proposed a grouped pixel embedding module to strengthen the feature extraction capability specifically for local spectral channels. Very recently, our previous work, local Transformer with spatial partition restore network (SPRLT-Net) was proposed in [35], where the SPR module is used to convert the HSI patch to sequence form, and the local Transformer block is used to calculate spatial attention within the patch. It is worth noting that Transformer-based models used for HSI classification commonly treat each pixel within the HSI patch as an individual token [33], [34], [35]. This approach differs from other vision tasks, where several image patches serve as tokens. For example, ViT employs a 14×14 tokenization strategy. Therefore, HSI classification still necessitates feature learning on a substantial number of tokens. Furthermore, HSI data typically possess dozens or even hundreds of times more channels than RGB images, resulting in the potential for features to be mapped to higher dimensions within the model. Consequently, Transformer-based models in HSI classification continue to confront the challenge of managing a large number of parameters.

However, labeled samples are extremely limited for HSI classification since manual labeling is time-consuming and expensive. At the same time, many classic DL frameworks struggle to deliver satisfactory performance when trained with limited samples. Previously, Shell et al. [62] pointed out that deep models are prone to overfitting issues when training data are severely constrained. To tackle this challenge, a prototypical network was designed to learn a metric space for classification, even when provided with only a few training samples for each class. Soon after, in order to enable the model to quickly adapt to a new task with only a small number of samples, Finn et al. [63] proposed a model-agnostic metalearning algorithm. They evaluated the performance of this method using five, ten, and 20 training samples per class, respectively. Subsequently, Sung et al. [64] employed a relation network as the architecture, implemented a metalearning strategy for model training, and conducted few-shot experiments under one- and five-shot settings. In the field of HSI classification, various effective methods have been investigated to address this challenge, such as transfer learning [36], [37], [38], [39], [40], active learning [41], [42], [43], [44], and few-shot learning (FSL) [45]. Among these studies, Liu et al. [59] earlier trained a deep CNN model using increments of five samples per class, ranging from 5 to 25, for few-shot HSI classification. Jia et al. [45] conducted a systematic literature review on few-shot HSI classification and compared the classification accuracy of eight advanced methods across three datasets with ten training samples per class. Generally, if the number of training samples for each category is significantly smaller than the number of HSI bands, the issue can be considered an FSL problem. At present, FSL methods, mainly including the prototype network, the relation network, and the Siamese network, have been extensively studied for HSI classification.

1) The **prototype network** believes that each category can be represented by a prototype vector, and samples clustered around the same class prototype in the feature space belong to the same category. In [46], hybrid residual attention is used to build a prototypical network, which can strengthen the ability of the model to mine spectral–spatial information. Subsequently, in order to reduce feature distribution bias between training and test sets, Wang et al. [47] introduced adversarial learning to prototype networks. Recently, a multiview calibrated framework based on prototype learning was proposed to improve the generalization ability of prototypes in feature space [48].
2) The **relation network** consists of an embedding module and a relation module. The former is used to extract the features of samples, and the latter adopts a learnable metric function to measure the similarity between features. To realize general information sharing between different datasets, Ma et al. [49] designed a two-phase model that consists of a relation learning method and an episode-based training strategy. Then, a spatial–spectral relation network, which can measure the similarity between samples, was proposed in [50] for HSI classification with limited training samples. Besides, a modified relation network based on depthwise separable convolution was proposed in [51] to reduce the computational cost of the model. However, previous methods all adopted a single metric function, which cannot effectively reflect the category distribution with limited training samples. To solve this issue, our recent work proposed a dual-metric method for few-shot HSI classification in [52].
3) The **Siamese network** consists of two weight-sharing subnetworks and a similarity measurement module. These two components work collaboratively to extract features from the input sample pair and distinguish whether the input pair belongs to the same category. Wang et al. [53] introduced spectral attention into the Siamese network earlier to extract discriminative spectral features from HSI cubes with different patch sizes. Later, Jia et al. [54] inserted the Siamese network into a semisupervised learning scheme, allowing the model to perform robustly under the condition of limited labeled samples. In addition, our recent work, spectral–spatial Siamese network (S3Net), was proposed in [55] to resolve the challenge of overfitting in DL models when training samples are limited. Similarly, we further proposed a differentiated-scale restricted GCN (DSR-GCN) method for HSI classification, where S3Net was used to remedy the local information [61].

Although the above DL methods have achieved excellent performance in HSI classification, there are still several limitations.

1) *Transformer for HSI Classification:* Transformer-based models utilized in HSI classification commonly employ a relatively fixed architecture, comprising multihead self-attention (MHSA), channel multilayer perceptron (MLP), and residual connection. Such architectural

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design can strengthen the model's capability to extract global features from the HSI patch. However, it not only results in a significant number of model parameters but also overlooks certain distinctive properties inherent to HSI, such as central symmetry and mixed pixels.

2) *FSL:* Most of the current FSL methods heavily rely on complex training techniques, such as metric learning and data augmentation, as well as sophisticated model structures, such as dual-stream networks and multiple loss functions. While these methods improve accuracy by expanding training samples or accounting for special cases ignored by traditional DL methods, they also introduce additional complexity. Nevertheless, there is limited research exploring simplified architectures for few-shot HSI classification.
3) *Transformer for Few-Shot HSI Classification:* Transformer discards some inductive biases, which requires a substantial number of training samples for compensation. Consequently, previous research heavily relies on CNN architecture for few-shot HSI classification. There is a scarcity of studies exploring methods using pure Transformer architecture.

Very recently, the general architecture of the Transformer called MetaFormer is proposed in [56] to replace the token mixer module in the Transformer with spatial pooling, which achieves better performance compared with a well-tuned Transformer-based model under the condition of reducing a large number of model parameters. Furthermore, MetaFormer argues that the token mixer module should extend beyond attention-based implementations, avoiding restriction to a specific type. In addition, improvements to the Transformer architecture model may not exclusively focus on this singular module.

In this article, we are inspired by MetaFormer and propose a lightweight self-pooling Transformer (SPFormer) for few-shot HSI classification. First, an autoencoder is used for dimensionality reduction, and its function is to alleviate the problem of spectral information redundancy and control the overall parameter amount of the model. Second, a channel shuffle for multihead self-pooling with sparse mapping (CSSM-MHSP) module is proposed to realize information flow between different channels and enable more distinguishable feature representation, and a central token mixer (CTM) module is proposed to facilitate information exchange between different pixels within the same patch. These two modules serve as spectral and spatial token mixers, respectively, within the Transformer architecture. Inspired by MetaFormer, this adaptation expands beyond the utilization of attention-based methods for implementing the token mixer module. Third, a lightweight channel embedding is designed to reintegrate spectral features, which simplifies the structure significantly compared to the channel MLP in Transformer and is well-suited for few-shot HSI classification. Finally, global average pooling (GAP) in conjunction with a fully connected layer is used for classification.

Briefly, the main contributions of our work can be summarized as follows.

1) We propose a novel lightweight Transformer architecture model, SPFormer, for few-shot HSI classification. Although techniques for expanding training samples such as data augmentation and common methods in the field of FSL such as metric learning are not used in our model, the proposed single-stream SPFormer trained with only one loss function can still show superiority on the four experimental datasets.
2) An autoencoder is trained using self-supervised learning to perform dimensionality reduction on HSI data. Compared with the commonly used PCA method, the features derived from the autoencoder preserve the most representative spectral information while mitigating the issue of substantial differences in information content across different components.
3) Two parameter-free modules, CSSM-MHSP and CTM, are proposed to serve as token mixer modules for spectral and spatial dimensions, respectively. These two modules replace the conventional MHSA module in traditional Transformer models.
4) We revisit the structure of channel MLP in Transformer and design channel embedding, which is more suitable for few-shot HSI classification. This design significantly reduces the amount of model parameters while ensuring its ability to effectively integrate representations across different spectral features even with limited training samples.

II. Proposed Methodology

Taking the University of Pavia dataset as an example, an overview of the proposed SPFormer is shown in Fig. 1, which is composed of three stages. In the first stage, an autoencoder is trained by each pixel in HSI, with the purpose of mapping the original data into a lower dimensional representation. In the second stage, the feature patch centered on the target pixel is fed into the backbone of SPFormer, which consists of two SPF blocks. Finally, GAP and a fully connected layer are used to classify the feature map output by the SPFormer backbone, and the classification result will serve as the predicted label for the center pixel of the original patch. Layer specification and the input and output sizes of each module in the proposed SPFormer are presented in Table I. Notations used in SPFormer are shown in Table II. To facilitate the description of dimension changes in the feature map, the patch size is set to 11×11, and the feature dimension output by autoencoder is set to 32.

A. HSI Dimensionality Reduction

In the case of limited training samples, without reducing the dimensionality of the data, training a DL model from scratch will greatly increase the model parameters, causing the model to fall into the problem of overfitting undoubtedly. Therefore, a dimensionality reduction module for few-shot HSI classification is necessary. In most of the prior studies, PCA [46], [48], [52], [55] has been predominantly employed for data dimensionality reduction. As shown in Fig. 1, the proposed method adopts a two-layer autoencoder for dimensionality reduction, and mean square error loss (MSELoss) is used for

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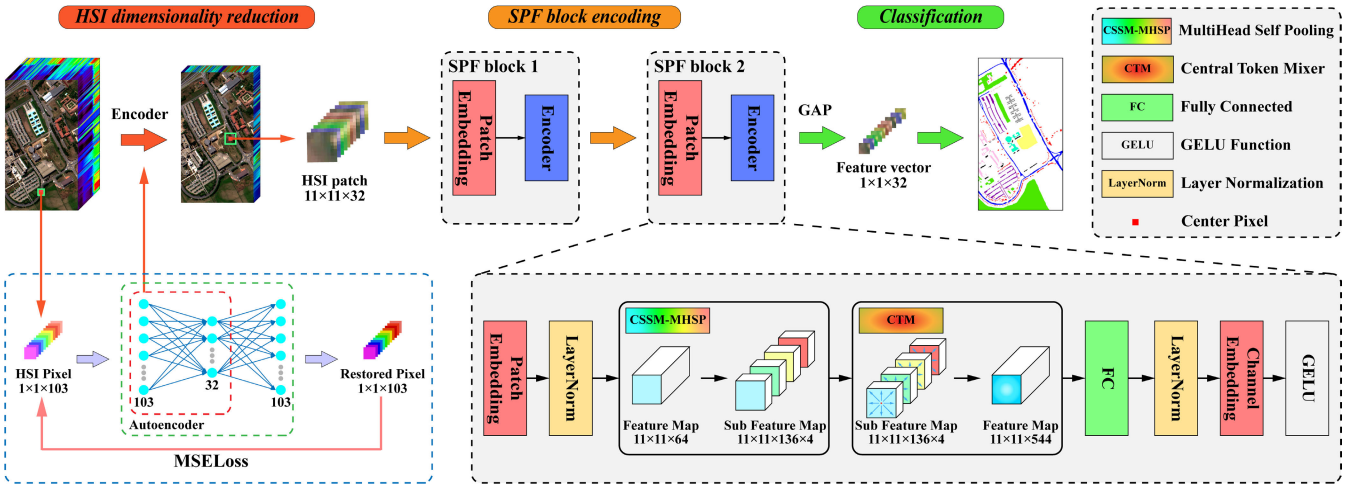


Fig. 1. Graphical illustration of the proposed SPFormer.

TABLE I

ARCHITECTURE DETAILS OF THE PROPOSED SPFORMER (TAKING THE UNIVERSITY OF PAVIA DATASET AS AN EXAMPLE)

Stage	Module	Input size	Layer specification	Output size
Autoencoder	Encoder head	(11,11,103)	Linear (dim=32)	(11,11,32)
SPF block×2	Patch embedding	(11,11,32)	Linear (dim=64)	(11,11,64)
	LayerNorm	(11,11,64)	LayerNorm (dim=64)	(11,11,64)
	CSSM-MHSP	(11,11,64)	Channel shuffle (groups=4) MHSP (head=4, headdim=16) Sparse mapping(256→136)	(11,11,136,4)
	CTM	(11,11,136,4)	Central token mixer	(11,11,544)
	FC	(11,11,544)	Linear (dim=32)	(11,11,32)
	LayerNorm	(11,11,32)	LayerNorm (dim=32)	(11,11,32)
	Channel embedding	(11,11,32)	Linear (dim=32)	(11,11,32)
	GELU	(11,11,32)	GELU activation function	(11,11,32)
Pooling	GAP	(11,11,32)	Global average pooling	(1,32)
Classification	FC	(1,32)	Linear (dim=9)	(1,9)

training, which can be represented as

$$L_{\text{mse}} = \frac{\sum_{n=1}^{N_1} \left\| x^{(n)} - \left[W_D \left(W_E x^{(n)} + b_E \right) \right] + b_D \right\|_2^2}{N_1} \quad (1)$$

where N_1 represents the total number of pixels in HSI; $x^{(n)}$ represents the spectral vector of the nth pixel; and W_E, W_D, b_E, and b_D represent the learnable weights and biases of the encoder and the decoder, respectively.

From our perspective, two reasons are given to explain the superiority of the autoencoder compared to PCA. For one thing, the result acquired from PCA does not contain equal information in different components. Autoencoder avoids this problem to some extent by using the learning mode. For another, PCA can be regarded as a linear transformation that only performs coordinate axis conversion. In contrast, the autoencoder goes beyond PCA by incorporating learnable parameter biases to learn the optimal offset of the coordinate origin. Moreover, both autoencoder and PCA methods utilize all HSI pixels as training samples. However, the distinction lies in their approaches. PCA conducts statistical learning on the entire data distribution of HSI, obtaining an analytical solution directly. Conversely, the autoencoder employs mini-batch gradient descent to optimize the model. Both methods can be considered unsupervised feature learning methods. Therefore, no label information is incorporated when training the autoencoder, implying that the autoencoder is only used for data dimensionality reduction and does not influence the classification results of the model.

In addition, in [65], an autoencoder was constructed based on MLP and CNN, employing unsupervised training for HSI dimensionality reduction, which is similar to our work. The main difference between this study and the autoencoder employed in SPFormer lies in whether to use the nonlinear function to perform feature transformation on HSI. The MLP and CNN autoencoder in [65] utilize activation function for nonlinear mapping on the original HSI, while the autoencoder employed in SPFormer adopts a more concise linear mapping strategy.

B. SPF Block

1) CSSM-MHSP Module: As shown in Fig. 2, the core module of SPFormer, CSSM-MHSP, consists of three submodules: channel shuffle (CS), multihead self-pooling (MHSP), and sparse mapping (SM).

1) *Channel Shuffle:* CS, which allows channel information to flow between different groups without increasing any parameters, was first proposed in ShuffleNet [57]. We think that shuffling the order of channels enhances the randomness during the training process and promotes diverse expression of sample features, ensuring the model not to be too sensitive to a specific feature representation and improving the robustness of the

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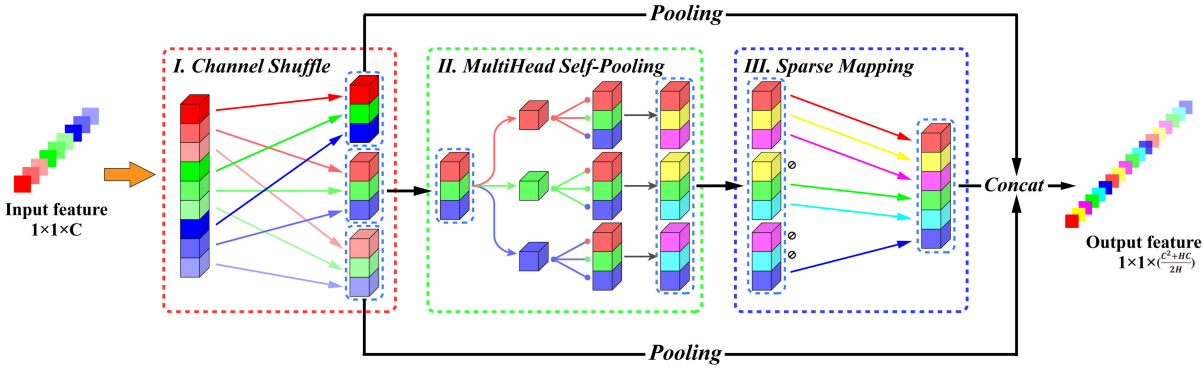


Fig. 2. Graphical illustration of the proposed CSSM-MHSP module.

TABLE II
NOTATIONS USED IN SPFORMER

Stage	Notation	Definition
Autoencoder	N_1	Number of pixels in HSI
	$x^{(n)}$	The nth pixel in HSI
	W_E	The weights of encoder in autoencoder
	b_E	The biases of encoder in autoencoder
	W_D	The weights of decoder in autoencoder
	b_D	The biases of decoder in autoencoder
	L_{MSE}	Mean square error loss function
SPF block	G	Number of groups in channel shuffle
	H	Number of heads in multi-head self-pooling
	M	Sparse matrix in sparse mapping
	x	Spectral feature of HSI pixel
	x_n	The nth value in spectral feature of HSI pixel
	x^k	Sub-feature of the kth head in HSI pixel
	X^k	Feature obtained by mapping sub-feature x^k
	X'	Spectral feature obtained by concatenating X^k $(k = 1, 2, \cdots, H)$ along channel dimension
	F	Feature map composed of all HSI pixel features
	F^k	Sub-feature map of the kth head in HSI feature map
	F_i^k	Feature of the ith pixel in the sub-feature map F^k
	F_{mid}^k	Features of the central pixel in the sub-feature map F^k
	ρ_i^k	Cosine similarity between F_i^k and F_{mid}^k
	ω_i^k	Attention score between F_i^k and F_{mid}^k
	Ω^k	Attention vector composed of ω_i^k in the kth sub-feature map
Classification	N_2	Number of training samples
	$y^{(i)}$	True label of ith sample
	$\hat{y}^{(i)}$	Predicted class probabilities of ith sample
	L_{CE}	Cross entropy loss function

model for few-shot HSI classification. The pseudocode of the CS algorithm is shown in Algorithm 1.

2) *Multihead Self-Pooling:* Inspired by MHSA in Transformer and spatial pooling in MetaFormer, we use spectral pooling to replace the learnable QKV linear projection, which maps spectral features to a higher dimension and effectively controls the number of model parameters. Assuming that the spectral features of an HSI pixel can be represented by a vector $x = (x_1, x_2, \ldots, x_C) \in \mathbb{R}^C$, an outer product of vector x on itself is conducted to obtain matrix $X \in \mathbb{R}^{C\times C}$. Then, matrix X is flattened into a vector form as the feature of vector x, and we define this operation as self-pooling,

Algorithm 1 CS

Input: Feature map $X \in \mathbb{R}^{B\times C\times H\times W}$, and number of groups G to divide along channel dimension.
Output: Feature map $X^* \in \mathbb{R}^{B\times C\times H\times W}$.
1 Reshape the dimension of X into $\left(B, G, \frac{C}{G}, H, W\right)$ to get $X \in \mathbb{R}^{B\times G\times \frac{C}{G}\times H\times W}$;
2 Swap the order of the second dimension (groups) and the third dimension (channels) of X;
3 Merge the second and the third dimension of X to get $X^* \in \mathbb{R}^{B\times C\times H\times W}$.

which can be formulated as

$$X = x \otimes x = \begin{bmatrix} x_1x_1 & x_1x_2 & \cdots & x_1x_C \\ x_2x_1 & x_2x_2 & \cdots & x_2x_C \\ \vdots & \vdots & \ddots & \vdots \\ x_Cx_1 & x_Cx_2 & \cdots & x_Cx_C \end{bmatrix} \quad (2)$$

$$\text{Self-pooling}(x) = (x_1x_1, x_1x_2, \ldots, x_Cx_C) \in \mathbb{R}^{C^2} \quad (3)$$

where $\otimes$ represents the outer product operation. Considering that the feature dimension obtained by self-pooling is high, the multihead strategy is innovatively used for feature dimensionality reduction. Specifically, vector $x \in \mathbb{R}^C$ is equally divided into H subvectors, and a self-pooling operation is performed on each subvector, which can be expressed by the following formula:

$$x^k = \left(x_{\frac{(k-1)C}{H}+1}, x_{\frac{(k-1)C}{H}+2}, \ldots, x_{\frac{kC}{H}}\right) \in \mathbb{R}^{\frac{C}{H}} \quad (4)$$

$$X^k = \text{Self-pooling}\left(x^k\right) \in \mathbb{R}^{\frac{C^2}{H^2}} \quad (5)$$

where x^k represents the kth $(k = 1, 2, \ldots, H)$ subfeature. At the same time, the parameter H in the MHSP module is set to the same value as G in the CS module in the proposed SPF block. Finally, all subfeatures are sequentially spliced as MHSP features

$$X' = \text{Concat}\left(X^1, X^2, \ldots, X^H\right) \in \mathbb{R}^{\frac{C^2}{H}}. \quad (6)$$

Furthermore, we think that implementing MHSP on the spectral features following the CS module can enhance feature diversity during the feature mapping process.

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Without the CS module, MHSP can only perform feature mapping on specific sets of spectral features in a fixed order. Using the CS module to reorganize the spectral features ensures that the model conducts the information flowing among various sets of spectral features before performing MHSP for feature mapping. This operation allows the model to dynamically map features across the entire spectral features of HSI, rather than being limited to only adjacent spectral features.

3) *SM:* For each subfeature $X^k \in \mathbb{R}^{(C^2/H^2)}$ obtained by MHSP module, it is obtained through flattening symmetric matrix operation. However, the lower triangular elements are redundant for feature representation. Therefore, in the CSSM-MHSP module, only the upper triangular element in the symmetric matrix acquired from outer product operation is reserved as the feature of the input patch. To achieve the above operations, we left-multiply a full-rank sparse matrix M to each subfeature for redundancy removal and ensure that this module is differentiable. For $1 \le i \le j \le (C/H)$, the sparse matrix M and the algorithm of the SM method can be represented as the following formula:

$$M\left[\frac{(i-1)C}{H}+j,\ \frac{(i-1)\left(\frac{2C}{H}-i\right)}{2}+j\right]=1 \tag{7}$$

$$X^k = MX^k \in \mathbb{R}^{\frac{C^2+HC}{2H^2}} \tag{8}$$

$$X' = \text{Concat}\left(X^1, X^2, \ldots, X^H\right) \in \mathbb{R}^{\frac{C^2+HC}{2H}} \tag{9}$$

where only the elements at specific positions are 1 and the elements at other positions are 0 in matrix M.

2) CTM Module: HSI data have a serious problem of mixed pixels. Considering that the edge pixels in the patch may have different land cover information from the central pixel, which is not conducive to classification, self-attention is not strictly used in the SPF block. Instead, it focuses on computing the attention between the central pixel and its surrounding pixels. Therefore, a parameter-free attention module, CTM, is used to realize information interactions between different pixels in each subfeature map, the details of which are described as follows.

Since the CSSM-MHSP module outputs the feature of each pixel $X' \in R^{((C^2+HC)/2H)}$, let F be the feature map composed of all pixel features in the patch. Assuming that $F^k \in \mathbb{R}^{H\times W\times((C^2+HC)/2H^2)}(k = 1, 2, \ldots, H)$ is the kth subfeature map obtained by multihead strategy, $F^k_{\text{mid}} \in \mathbb{R}^{1\times1\times((C^2+HC)/2H^2)}$ is the feature of center pixel in the kth subfeature map. As shown in Fig. 3, the operation process of the CTM module can be decomposed into the following three steps.

First, F^k is expanded into a 2-D matrix of $HW \times ((C^2+HC)/2H^2)$ dimension. For the convenience of subsequent description, we record $N = HW$, i.e., $F^k \in \mathbb{R}^{N\times((C^2+HC)/2H^2)}$. Note that F^k_i is feature of the ith $(i = 1, 2, \ldots, N)$ pixel in the sequence. Second, $L2$ normalization is applied to F^k_i along the channel dimension, and cosine similarity ρ^k_i is computed between F^k_{mid} and other pixels F^k_i within the sequence. Based on ρ^k_i, attention score ω^k_i is then calculated using the softmax function. The above operation can be represented as the following formula:

$$\rho^k_i = \frac{\left(F^k_i\right)\left(F^k_{\text{mid}}\right)^T}{\left\|F^k_i\right\|\left\|F^k_{\text{mid}}\right\|} \tag{10}$$

$$\omega^k_i = \frac{e^{\rho^k_i}}{\sum_{i=1}^{N} e^{\rho^k_i}}. \tag{11}$$

Finally, arrange ω^k_i into a vector $\Omega^k \in \mathbb{R}^{N\times1}$ in order, make the elementwise product of Ω^k and subfeature map F^k, and fold each subfeature map F^k back into the form of the cube. In addition, considering $\sum_{i=1}^{N}\omega^k_i = 1$, to maintain the magnitude of F^k unchanged before and after the operation, a constant N is multiplied on the left side of the equation. Finally, all $F^k \in R^{H\times W\times((C^2+HC)/2H^2)}(k = 1, 2, \ldots, H)$ are stacked along channel dimension to get $F \in R^{H\times W\times((C^2+HC)/2H)}$. The formulaic expression of the above operation is given as follows:

$$\Omega^k = \left(\omega^k_1, \omega^k_2, \ldots, \omega^k_N\right) \tag{12}$$

$$F^k = N\Omega^k \odot F^k \tag{13}$$

$$F = \text{Concat}\left(F^1, F^2, \ldots, F^H\right) \tag{14}$$

where $\odot$ represents the elementwise product.

3) Channel Embedding With Gaussian Error Linear Units (GELU) Activation Function: At the end of the SPF block, a channel embedding with fewer parameters, mapped to the same dimension and connected with a GELU activation function, facilitates the feature integration of subfeatures from different heads. The expression of the GELU activation function is shown in formula (15). Compared with channel MLP in the traditional Transformer model, which consists of two fully connected layers and one activation function, the architecture of the corresponding module in the SPF block significantly reduces the number of model parameters, ensuring easy convergence of the model even with limited training samples

$$\text{GELU}(x) = 0.5x\left(1+\text{Tanh}\left(\sqrt{\frac{2}{\pi}}\left(x+0.044715x^3\right)\right)\right) \tag{15}$$

where x represents the input feature map.

C. Classification

For the 3-D feature map output by the SPF block, GAP is used to transform it into a 1-D vector form as the feature of the HSI patch, and a fully connected layer is used for classification. Finally, the cross-entropy loss function is used to calculate training loss and update model weights, which can be represented as

$$L_{\text{CE}} = \frac{-\sum_{i=1}^{N_2} y^{(i)} \log\left(\hat{y}^{(i)}\right)}{N_2} \tag{16}$$

where N_2 is the total number of training samples, and $y^{(i)}$ and $\hat{y}^{(i)}$ are the true label and the predicted class probabilities of the ith sample, respectively.

At the end of this section, the pseudocode of the SPFormer algorithm is shown in Algorithm 2.

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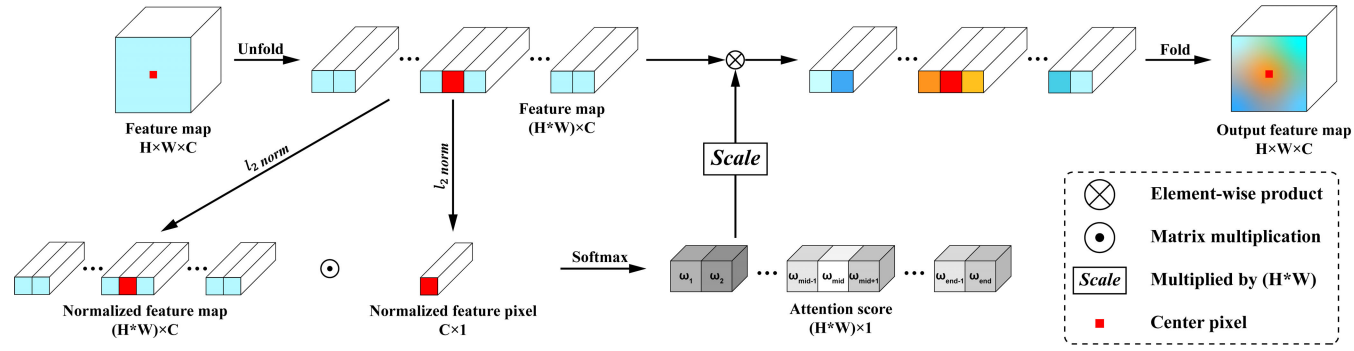


Fig. 3. Graphical illustration of the proposed CTM module.

Algorithm 2 SPFormer

Input: HSI data $I \in \mathbb{R}^{m\times n\times C}$ and ground truth image $Y \in \mathbb{R}^{m\times n}$, reduce dimension c, training epoch for autoencoder $Epo1$, training epoch for SPFormer $Epo2$.

Output: Predicted labels for HSI.

I. Autoencoder Training Stage:
for *i=1 to Epo1* **do**
  Feed each spectral vector V_{raw} from I into autoencoder to get restored vector $V_{restore}$;
  Calculate L_{MSE} using Eq.(1) by V_{raw} and $V_{restore}$;
  Train autoencoder using L_{MSE} with Adam optimizer.
end
II. SPFormer Training Stage:
Obtain dimension-reduced data $F_{AE} \in \mathbb{R}^{m\times n\times c}$ using autoencoder;
Split labeled data into trainset S_{train}: $\{x_{train}, y_{train}\}$ and testset S_{test}: $\{x_{test}, y_{test}\}$;
for *i=1 to Epo2* **do**
  Feed each sample x_{train} from S_{train} into SPFormer to get feature vector $\hat{y}_{train}$;
  Calculate L_{CE} using Eq.(16) by y_{train} and y_{test};
  Train SPFormer using L_{CE} with Adam optimizer.
end
III. Test Stage:
Feed samples x_{test} from S_{test} into trained SPFormer to get predicted labels.

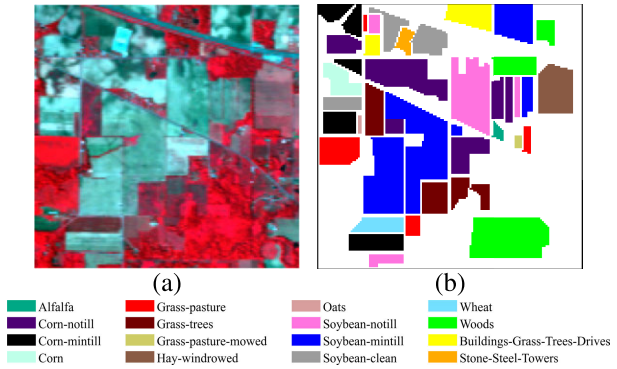


Fig. 4. Indian Pines. (a) False-color composite image (R: 50, G: 27, B: 17). (b) Ground-truth map.

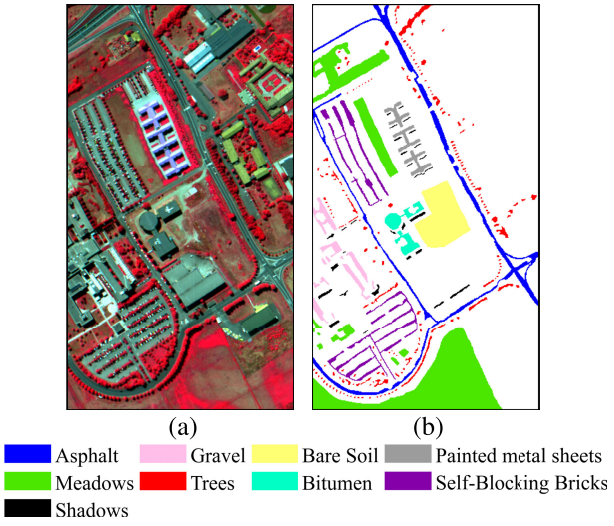


Fig. 5. University of Pavia. (a) False-color composite image (R: 102, G: 56, B: 31). (b) Ground-truth map.

III. Experimental Results

A. Hyperspectral Datasets

Four widely used HSI datasets, including Indian Pines, the University of Pavia, Salinas, and the University of Houston, are employed to conduct the experiments. The size and spectral bands of these datasets are given as follows: Indian Pines: $\{(145 \times 145), 224\}$; University of Pavia: $\{(610 \times 340), 103\}$; Salinas: $\{(512 \times 217), 224\}$; and University of Houston: $\{(349 \times 1905), 144\}$. In addition, bands damaged by water vapor absorption in Indian Pines and Salinas datasets are excluded. The three-band false-color composite images and ground-truth maps corresponding to these four datasets are shown in Figs. 4–7, respectively. The number of labeled samples of different categories in each dataset is recorded in Table III.

B. Experimental Settings

1) Hyperparameters: Exponential decay is used as a learning rate decay trick, and the decay rate is set to 0.99. For the training epoch, the training process would be stopped once the training loss is stabilized. Specifically, the training epoch is set to 160 for Indian Pines, 150 for the University of Pavia, 170 for

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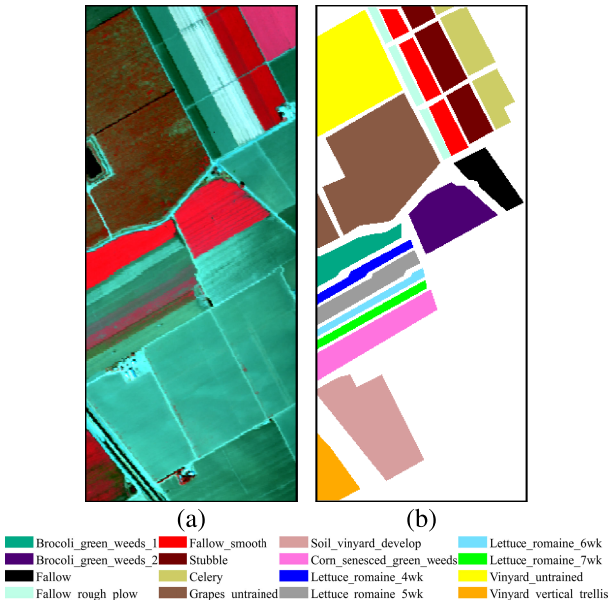


Fig. 6. Salinas. (a) False-color composite image (R: 57, G: 27, B: 17). (b) Ground-truth map.

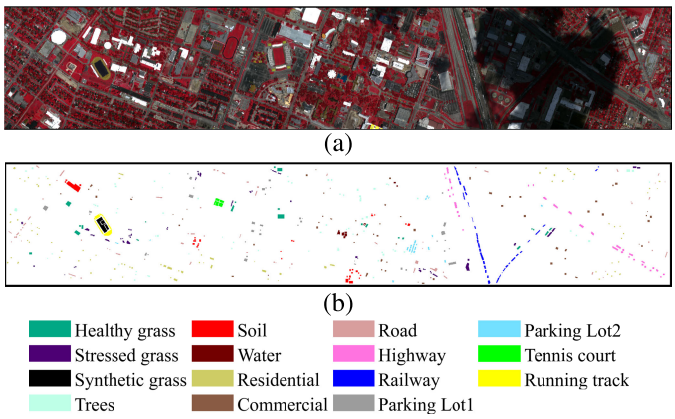


Fig. 7. University of Houston. (a) False-color composite image (R: 59, G: 40, B: 23). (b) Ground-truth map.

Salinas, and 220 for the University of Houston. The batch size is set to 80 for Indian Pines, 45 for the University of Pavia, 80 for Salinas, and 75 for the University of Houston. Other parameters, including learning rate, patch size, data dimensionality reduction, and the number of heads in the CSSM-MHSP module, will be optimized in the following experiments.

2) Training Samples: We randomly select [5, 10, 15, 20, 25] samples from each class in each experimental dataset to construct a training set, and the remaining samples are constructed as the test set for classification accuracy evaluation. In addition, make sure that the number of samples in the training set is not more than that of the test set.

3) Performance Comparison: Some representative methods are compared to verify the effectiveness of SPFormer for hyperspectral FSL, including two typical deep models (3-D CNN (3DCNN) [58] and spectral-spatial residual network (SSRN) [17]), three transfer learning methods (deep few-shot learning - NN (DFSL-NN) [59], deep few-shot learning - SVM (DFSL-SVM) [59], and deep cross-domain few-shot learning (DCFSL) [38]), one prototype network model (deep prototypical network with hybrid residual attention (DPN-HRA) [46]), one relation network model (deep relation network for HSI few-shot classification (RN-FSC) [60]), two Siamese network models (a semisupervised Siamese network with an autoencoder module (3DAES) [54] and spectral-spatial Siamese network (S3Net) [55]), and two state-of-the-art Transformer-based models (spectral-spatial feature tokenization Transformer (SSFTT) [33] and group-aware hierarchical Transformer (GAHT) [34]). The parameters of all the above comparison methods are set to be the same as those in the original article.

4) Running Platforms and Metrics: All experiments were run with Intel[1] Xeon[1] CPU, GeForce RTX 3060 (12 GB), and 32-GB RAM based on PyTorch 1.10.2 and cuDNN 11.1.1 framework. The results of ten independent experiments are averaged as the results are presented. Three commonly used metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), are used to evaluate the classification accuracy of the model on the test set. Besides, training time (s), test time (s), and the amount of model parameters are used to evaluate the model complexity. The source code will be available online at https://github.com/ZhaohuiXue/SPFormer.

C. Parameter Sensitive Analysis

1) Learning Rate: To determine the suitable learning rate for model training, we sequentially set the learning rate from the range of $[1e-1, 5e-2, 1e-2, 5e-3, 1e-3, 5e-4, 1e-4]$. Subsequently, we conduct experiments on the aforementioned four datasets and record the OA achieved by the model under each learning rate configuration. As shown in Fig. 8(a), when the learning rate ranges from $1e-1$ to $1e-3$, the OA obtained by SPFormer exhibits a consistent upward trend across four datasets, reaching its peak when the learning rate is set to $1e-3$. However, OA shows an obvious decline when the learning rate falls below $1e-3$. Based on the experimental results, we set the learning rate to $1e-3$ for subsequent experiments.

2) Patch Size: In this experiment, the impact of patch size on classification accuracy is evaluated. We set the patch size to [5, 7, 9, 11, 13, 15, 17], respectively. As shown in Fig. 8(b), when the patch size is set to 13×13, the highest OA can be achieved compared to other configurations for Indian Pines and Salinas datasets. For the University of Pavia and University of Houston datasets, the highest OA can be achieved when the patch size is set to 11×11. Based on the experimental results, we set the patch size to 13×13 for Indian pines and Salinas and 11×11 for the University of Pavia, respectively.

3) Dimensionality Reduction Using Autoencoder: The original HSI data are reduced to [24, 28, 32, 36, 40, 44, 48] dimensions using autoencoder. These dimensionality-reduced features are then fed into the SPFormer backbone, and the resulting impact on classification accuracy is analyzed across different feature dimensions. As shown in Fig. 8(c), SPFormer obtains higher classification accuracy than other parameter configurations when HSI data are reduced to 40 for Indian

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TABLE III
NUMBER OF LABELED SAMPLES PER CLASS FOR THREE DATASETS

Class	Indian Pines		University of Pavia		Salinas		University of Houston	
	Name	Number	Name	Number	Name	Number	Name	Number
1	Alfalfa	46	Asphalt	6631	Brocoli green weeds 1	2009	Healthy grass	1251
2	Corn-notill	1428	Meadows	18649	Brocoli green weeds 2	3726	Stressed grass	1254
3	Corn-mintill	830	Gravel	2099	Fallow	1976	Synthetic grass	697
4	Corn	237	Trees	3064	Fallow rough plow	1394	Trees	1244
5	Grass-pasture	483	Painted metal sheets	1345	Fallow smooth	2678	Soil	1242
6	Grass-trees	730	Bare soil	5029	Stubble	3959	Water	325
7	Grass-pasture-moved	28	Bitumen	1330	Celery	3579	Residential	1268
8	Hay-windrowed	478	Self-Blocking Bricks	3682	Grapes untrained	11271	Commercial	1244
9	Oats	20	Shadows	947	Soil vinyard develop	6203	Road	1252
10	Soybeans-notill	972			Corn senesced green weeds	3278	Highway	1227
11	Soybeans-mintill	2455			Lettuce romaine 4wk	1068	Railway	1235
12	Soybeans-clean	593			Lettuce romaine 5wk	1927	Parking Lot1	1233
13	Wheat	205			Lettuce romaine 6wk	916	Parking Lot2	469
14	Woods	1265			Lettuce romaine 7wk	1070	Tennis court	428
15	Bldg-grass-tree-drivers	386			Vinyard untrained	7268	Running track	660
16	Stone-streel-towers	93			Vinyard vertical trellis	1807		
	Total	10249	Total	42776	Total	54129	Total	15029

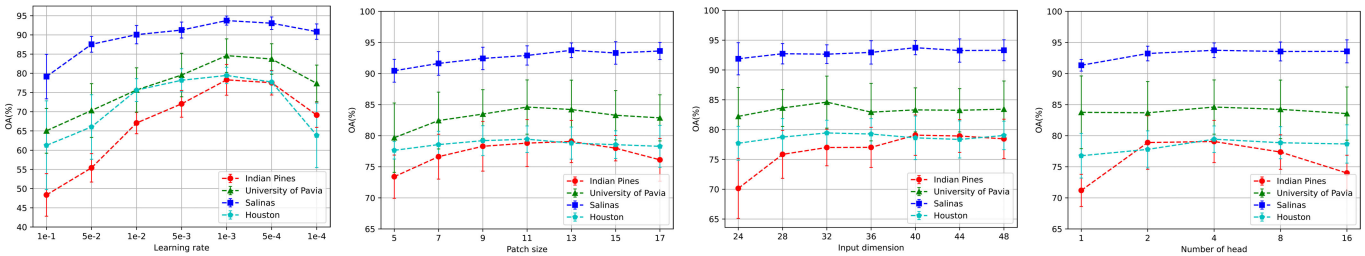


Fig. 8. Evolution of OA as a function of (a) learning rate, (b) patch size, (c) input dimension, and (d) number of head.

TABLE IV
ABLATION STUDY OF AUTOENCODER MODULE IN FOUR DATASETS

Datasets	Evaluation	MLP autoencoder [65]	CNN autoencoder [65]	Linear autoencoder
Indian Pines	OA (%)	59.58±3.33	58.73±2.80	**79.07±3.39**
	AA (%)	72.38±1.14	69.04±0.97	**87.78±1.90**
	$\kappa\times100$	54.81±3.28	53.48±2.75	**76.38±3.74**
	Training time (s)	14.12	37.21	7.37
University of Pavia	OA (%)	77.23±6.00	78.26±4.88	**84.61±4.36**
	AA (%)	85.61±3.10	84.25±2.66	**90.94±1.66**
	$\kappa\times100$	71.38±6.32	72.48±5.35	**80.67±5.07**
	Training time (s)	641.86	1779.34	320.52
Salinas	OA (%)	85.86±2.98	84.27±2.05	**93.74±1.18**
	AA (%)	89.12±1.72	85.58±1.31	**96.74±0.75**
	$\kappa\times100$	84.34±3.26	82.50±2.23	**93.05±1.31**
	Training time (s)	77.44	414.14	40.72
University of Houston	OA (%)	71.37±2.19	71.45±1.95	**79.44±2.13**
	AA (%)	74.86±2.50	75.84±1.91	**82.15±1.99**
	$\kappa\times100$	69.17±2.37	69.25±2.09	**77.81±2.30**
	Training time (s)	1149.26	3161.36	564.77

pines and Salinas and 32 for the University of Pavia and the University of Houston. Therefore, the above configurations are used for subsequent experiments.

4) Number of Head in SPF Block: This parameter affects the dimension of feature output by the intermediate layer and the parameter amount of the overall model. We set this parameter to [1, 2, 4, 8, 16], respectively, and compare the classification performance of the model under different settings. As shown in Fig. 8(d), SPFormer achieves higher classification accuracy than other configurations on the four experimental datasets when this parameter is set to 4.

D. Ablation Study

1) Autoencoder for Data Dimensionality Reduction: In this experiment, we compare the classification accuracy achieved by different dimensionality reduction methods, including the widely employed PCA method in few-shot HSI classification, along with three autoencoder methods: MLP autoencoder [65], CNN autoencoder [65], and linear autoencoder implemented in SPFormer. The results, presented in Table IV, outline the classification accuracy achieved by different autoencoders for HSI dimensionality reduction when integrated into the SPFormer backbone, along with the corresponding training time. Notably, linear autoencoder exhibits the shortest training time and achieves the highest classification accuracy. We believe that employing linear mapping in constructing the autoencoder preserves more information from HSI. In addition, the classification results obtained using the PCA method and linear autoencoder are reported in Table V. OA, AA, and $\kappa\times100$ obtained by the autoencoder method on four datasets have all been improved compared with the dimensionality reduction method based on PCA. Specifically, OAs increase by 1.19% for Indian Pines, 2.26% for the University of Pavia, 1.02% for Salinas, and 0.95% for the University of Houston, respectively.

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TABLE V
ABLATION STUDY OF SPFORMER MODULES IN FOUR DATASETS

Datasets	Dimension reduction		CSSM-MHSP			CTM	OA (%)	AA (%)	$\kappa\times100$
	PCA	Autoencoder	CS	MHSP	SM				
Indian Pines	✓	✗	✓	✓	✓	✓	77.88±2.94	86.97±1.61	75.06±3.25
	✗	✓	✗	✗	✗	✓	71.20±2.59	82.93±1.26	67.62±2.84
	✗	✓	✗	✓	✓	✓	77.57±3.65	86.82±2.00	74.75±4.01
	✗	✓	✓	✗	✓	✓	77.00±3.61	85.93±1.84	74.17±3.97
	✗	✓	✓	✓	✗	✓	**79.09±3.21**	87.63±1.87	76.37±3.58
	✗	✓	✓	✓	✓	✗	76.92±2.93	85.99±2.11	74.07±3.27
	✗	✓	✓	✓	✓	✓	79.07±3.39	**87.78±1.90**	**76.38±3.74**
University of Pavia	✓	✗	✓	✓	✓	✓	82.35±4.73	89.65±2.23	77.89±5.28
	✗	✓	✗	✗	✗	✓	83.76±5.85	88.60±2.39	79.47±6.85
	✗	✓	✗	✓	✓	✓	82.72±3.90	90.07±2.42	78.38±4.52
	✗	✓	✓	✗	✓	✓	82.59±3.59	89.58±1.92	78.22±4.05
	✗	✓	✓	✓	✗	✓	83.41±3.54	90.69±1.81	79.22±4.10
	✗	✓	✓	✓	✓	✗	82.78±5.04	89.62±2.01	78.49±5.75
	✗	✓	✓	✓	✓	✓	**84.61±4.36**	**90.94±1.66**	**80.67±5.07**
Salinas	✓	✗	✓	✓	✓	✓	92.72±1.97	95.94±1.39	91.93±2.17
	✗	✓	✗	✗	✗	✓	91.33±0.94	95.34±0.87	90.37±1.03
	✗	✓	✗	✓	✓	✓	93.22±1.66	96.33±0.91	92.47±1.84
	✗	✓	✓	✗	✓	✓	92.74±1.78	95.60±1.10	91.93±1.97
	✗	✓	✓	✓	✗	✓	93.02±1.77	96.28±0.78	92.24±1.96
	✗	✓	✓	✓	✓	✗	92.72±1.77	95.58±1.15	91.91±1.96
	✗	✓	✓	✓	✓	✓	**93.74±1.18**	**96.74±0.75**	**93.05±1.31**
University of Houston	✓	✗	✓	✓	✓	✓	78.49±2.24	81.31±2.05	76.79±2.41
	✗	✓	✗	✗	✗	✓	79.18±3.65	81.69±2.95	77.53±3.93
	✗	✓	✗	✓	✓	✓	78.59±2.38	81.19±2.35	76.90±2.55
	✗	✓	✓	✗	✓	✓	76.78±3.59	79.79±2.87	74.95±3.86
	✗	✓	✓	✓	✗	✓	77.83±2.82	80.87±2.36	76.08±3.04
	✗	✓	✓	✓	✓	✗	76.58±3.42	79.65±2.88	74.74±3.69
	✗	✓	✓	✓	✓	✓	**79.44±2.13**	**82.15±1.99**	**77.81±2.30**

TABLE VI
ABLATION STUDY OF CTM MODULE IN FOUR DATASETS

Datasets	Evaluation	Conv	Pooling	Spatial attention	MHSA	CTM
Indian Pines	OA (%)	72.21±3.22	73.38±3.75	76.83±3.25	54.11±5.79	**79.07±3.39**
	AA (%)	83.45±1.73	83.72±1.94	86.12±1.62	70.73±3.42	**87.78±1.90**
	$\kappa\times100$	68.93±3.49	70.23±4.06	73.97±3.58	49.18±6.09	**76.38±3.74**
	Parameters	12788776	86296	86332	4319896	86296
University of Pavia	OA (%)	80.36±3.70	81.46±3.44	83.14±3.49	67.63±7.90	**84.61±4.36**
	AA (%)	88.10±1.47	87.24±2.25	89.61±2.01	73.13±5.80	**90.94±1.66**
	$\kappa\times100$	75.54±4.08	76.67±3.97	78.88±4.04	60.33±8.44	**80.67±5.07**
	Parameters	5373161	45225	45261	1820841	45225
Salinas	OA (%)	92.33±1.84	92.15±1.74	92.93±1.37	78.72±4.25	**93.74±1.18**
	AA (%)	95.21±1.12	95.04±1.33	95.92±0.74	82.76±3.39	**96.74±0.75**
	$\kappa\times100$	91.49±2.03	91.28±1.92	92.14±1.52	76.47±4.67	**93.05±1.31**
	Parameters	12788936	86456	86492	4320056	86456
University of Houston	OA (%)	78.27±1.92	77.63±2.55	76.67±3.45	63.95±3.28	**79.44±2.13**
	AA (%)	81.40±1.75	80.94±2.26	79.71±2.86	68.64±2.71	**82.15±1.99**
	$\kappa\times100$	76.55±2.07	75.87±2.75	74.83±3.72	61.09±3.55	**77.81±2.30**
	Parameters	5374671	46735	46771	1822351	46735

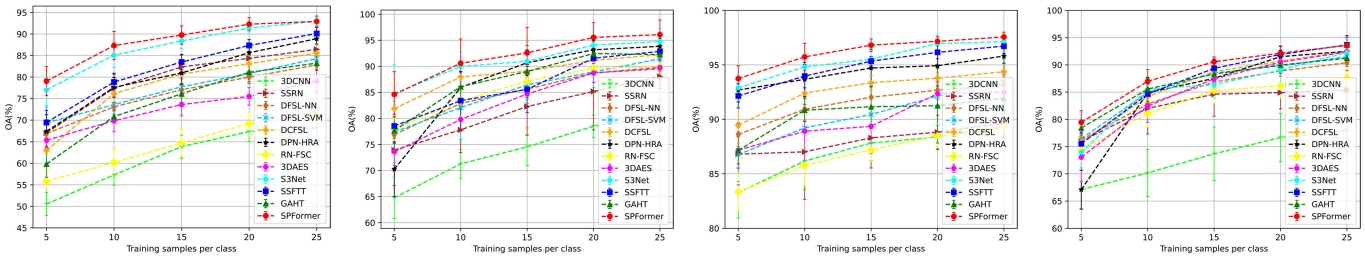
2) Effectiveness of CSSM-MHSP Module: The core of the Transformer architecture model is the self-attention module, which corresponds to CSSM-MHSP and CTM modules in the proposed SPFormer. In this part, we extract the CSSM-MHSP module and conduct separation experiments on each of its components, including CS, MHSP, and SM, to validate their effectiveness. As reported in Table V, using CS and MHSP can effectively improve classification accuracy compared with not using them. This is due to that CS can regularly reorganize the order of spectral features between groups, facilitating the circulation of spectral information between different groups and preventing the model from exclusively learning a single spectral pattern. MHSP can map the reorganized spectral features to a higher dimensional space, enhancing the separability of features. While the classification accuracy without using the SM strategy yields a slightly higher OA for the Indian Pines dataset, the difference is very small, and AA together with $\kappa \times 100$ obtained by using this strategy is higher. In addition, employed on datasets with more original bands such as Indian Pines and Salinas, the utilization of the CSSM-MHSP module leads to a substantial increase in accuracy. Specifically, OA improves by 7.87% for Indian Pines and 2.41% for Salinas, respectively.

3) Effectiveness of CTM Module: In the proposed SPFormer, the CTM module facilitates information interaction among tokens in the spatial dimension. To verify its effectiveness, we conduct separation experiments and substitute CTM with other similar methods. First, as reported in Table V, the utilization of the CTM module results in accuracy improvements across all four datasets. Taking OA as an example, it increased by 2.15% for Indian Pines, 1.83% for the University of Pavia, 1.02% for Salinas, and 2.86% for the University of Houston. Then, we substitute the CTM module with convolutional layer (Conv), spatial pooling (pooling), spatial

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TABLE VII
ABLATION STUDY OF CHANNEL EMBEDDING IN FOUR DATASETS

Datasets	Channel embedding	Channel MLP	Residual connection	OA (%)	AA (%)	$\kappa\times100$
Indian Pines	✗	✓	✓	72.61±3.26	83.82±1.50	69.16±3.51
	✓	✓	✗	76.97±3.96	86.37±2.42	74.01±4.37
	✓	✗	✓	72.24±3.52	83.44±1.74	68.69±3.86
	✓	✗	✗	**79.07±3.39**	**87.78±1.90**	**76.38±3.74**
University of Pavia	✗	✓	✓	78.50±4.15	87.54±2.28	73.14±4.50
	✓	✓	✗	84.13±3.34	90.36±2.31	79.99±3.89
	✓	✗	✓	78.88±4.28	87.79±3.36	73.56±4.86
	✓	✗	✗	**84.61±4.36**	**90.94±1.66**	**80.67±5.07**
Salinas	✗	✓	✓	91.94±1.57	95.28±0.98	91.05±1.72
	✓	✓	✗	93.28±1.42	96.04±1.03	92.53±1.57
	✓	✗	✓	92.14±2.16	95.15±1.46	91.28±2.38
	✓	✗	✗	**93.74±1.18**	**96.74±0.75**	**93.05±1.31**
University of Houston	✗	✓	✓	78.20±2.72	81.04±2.59	76.48±2.94
	✓	✓	✗	78.31±3.18	81.06±2.91	76.60±3.43
	✓	✗	✓	79.04±1.60	81.58±1.83	77.37±1.73
	✓	✗	✗	**79.44±2.13**	**82.15±1.99**	**77.81±2.30**



(a) (b) (c) (d)

Fig. 9. Evolution of OA as a function of training samples per class. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) Houston.

TABLE VIII
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN INDIAN PINES (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN [58]	SSRN [17]	DFSL-NN [59]	DFSL-SVM [59]	DCFSL [38]	DPN-HRA [54]	RN-FSC [60]	3DAES [54]	S3Net [55]	SSFTT [33]	GAHT [34]	**SPFormer (Ours)**
1	80.00±18.28	99.02±1.62	98.05±3.24	97.80±2.98	97.32±3.53	45.39±23.72	88.78±11.08	99.35±1.94	**100.00±0.00**	99.76±0.73	96.83±5.46	99.51±0.98
2	31.46±8.88	46.23±16.52	40.10±6.46	45.19±7.52	40.44±7.11	**75.96±14.55**	33.65±9.68	47.05±10.91	58.84±13.30	50.98±12.46	42.00±8.79	61.52±13.07
3	34.13±8.96	63.60±8.80	67.77±6.79	65.01±16.97	45.62±11.09	57.89±9.64	27.89±8.09	47.24±6.88	63.68±6.74	62.40±8.45	45.26±6.34	**71.25±9.58**
4	36.59±11.92	82.33±12.97	74.70±10.67	72.67±14.23	78.36±9.35	45.68±14.33	65.78±9.75	74.91±8.48	94.26±5.71	90.91±13.78	83.36±9.08	**97.50±3.15**
5	62.22±12.31	78.28±7.00	80.90±7.41	**83.54±8.95**	73.08±6.40	71.93±18.56	64.81±15.74	82.56±5.97	80.92±7.28	80.77±7.33	68.66±10.23	80.77±7.20
6	62.99±12.76	92.63±7.38	90.81±3.35	89.92±4.86	82.46±6.53	89.82±7.34	73.86±5.07	83.26±6.75	93.57±2.79	90.43±6.27	80.44±11.49	**94.63±3.06**
7	87.39±14.87	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	29.73±11.02	99.57±1.30	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
8	86.26±9.50	89.20±14.06	88.10±6.29	89.05±6.65	81.44±13.65	98.73±3.19	83.34±5.00	95.75±1.97	**99.87±0.27**	91.92±13.19	78.33±16.99	99.81±0.44
9	98.00±4.27	100.00±0.00	100.00±0.00	100.00±0.00	98.67±2.67	14.41±12.34	98.67±2.67	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
10	52.08±13.49	67.34±9.32	63.14±7.31	59.81±6.94	59.06±7.05	59.88±16.00	55.53±10.58	59.52±11.78	**71.04±7.72**	64.45±9.17	53.45±10.22	70.82±10.36
11	48.90±16.67	51.31±13.58	52.63±11.78	59.56±12.07	59.23±12.48	**76.15±9.89**	48.94±7.60	55.03±8.99	70.52±12.38	57.89±12.23	52.55±14.79	73.47±9.50
12	27.57±6.28	60.20±16.55	63.95±11.02	70.90±10.91	44.15±10.39	48.77±15.10	25.43±9.98	51.09±8.72	69.06±10.32	50.83±16.17	46.04±8.66	**76.90±20.20**
13	91.35±11.30	99.15±1.18	94.30±4.54	93.45±5.43	99.10±0.97	90.68±6.00	89.80±6.89	**99.68±0.67**	97.85±2.96	98.25±2.53	99.60±0.58	98.90±1.30
14	62.41±14.60	84.63±9.27	88.75±4.48	90.64±4.81	81.48±12.04	90.16±8.30	89.37±4.66	87.89±3.72	**94.84±7.10**	87.54±7.64	73.29±15.69	93.40±8.79
15	47.56±8.95	73.86±15.27	75.70±12.75	75.46±14.66	70.37±12.57	65.91±13.09	58.74±9.69	74.88±7.95	**88.50±10.89**	86.40±12.01	72.47±9.80	86.27±10.47
16	93.64±4.32	98.52±1.90	96.70±3.82	95.91±5.43	98.64±2.58	63.45±26.63	97.05±4.85	99.62±1.15	97.04±3.77	98.75±3.03	**99.89±0.34**	99.77±0.45
OA (%)	50.55±2.68	67.02±3.04	66.86±3.10	69.37±3.69	62.99±2.94	67.39±3.86	55.81±1.79	65.37±1.63	77.02±3.55	69.45±2.83	59.85±3.09	**79.07±3.39**
AA (%)	62.66±1.56	80.40±1.61	79.72±1.70	80.56±2.44	75.59±1.11	64.03±3.18	68.82±0.98	78.61±1.00	86.25±1.53	81.96±1.44	74.51±1.01	**87.78±1.90**
$\kappa\times100$	44.60±2.65	63.04±3.21	62.84±3.31	65.49±4.07	58.37±3.01	63.58±4.13	50.58±1.80	60.99±1.76	74.19±3.88	65.68±3.03	54.98±3.13	**76.38±3.74**

attention, and MHSA and conduct comparative experiments to validate the superiority of this module. As reported in Table VI, CTM achieves the highest accuracy among all methods, with fewer model parameters than convolutional layer, spatial attention, and multihead self-attention. Furthermore, compared to the parameter-free method, spatial pooling, the accuracy improvement that CTM achieves is significant.

4) Channel Embedding With GELU Activation Function: In addition to the self-attention module, another essential component of the Transformer architecture model is the feedforward network (FFN) module, which corresponds to the channel embedding module in the proposed SPFormer. In this part of the experiment, channel embedding and channel MLP are compared, and whether to use residual connection is explored to evaluate the influence of different model variants on the experimental classification accuracy results. According to Table VII, the classification accuracy achieved without utilizing the residual connection is notably higher than when it is used. Through comprehensive analysis of the impact of different structures on classification accuracy, we think that combining channel embedding with the GELU activation function in series can yield superior classification results.

E. Generalization Performance

The experimental results of different models for few-shot HSI classification are compared to evaluate the generalization performance of SPFormer in this experimental phase. We randomly select [5, 10, 15, 20, 25] samples from the four experimental datasets for training and the rest for testing. It can be indicated from Fig. 9 that SPFormer performs better than other models in most cases. For Indian Pines, Salinas,

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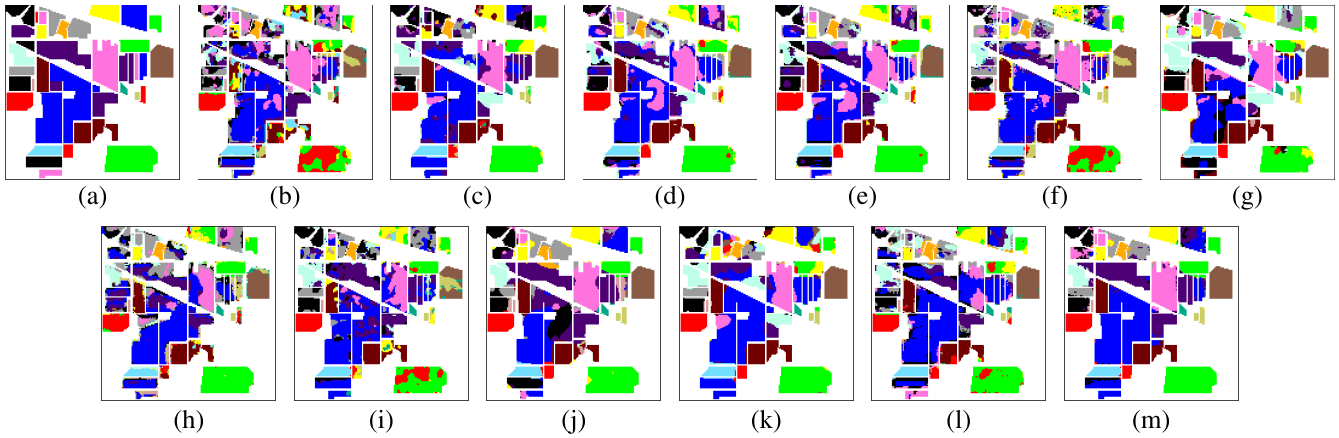


Fig. 10. Classification maps obtained by different methods in Indian Pines (five labeled samples per class for training). (a) Ground Truth. (b) 3DCNN (50.55%). (c) SSRN (67.02%). (d) DFSL-NN (66.86%). (e) DFSL-SVM (69.37%). (f) DCFSL (62.99%). (g) DPN-HRA (67.39%). (h) RN-FSC (55.81%). (i) 3DAES (65.37%). (j) S3Net (77.02%). (k) SSFTT (69.45%). (l) GAHT (59.85%). (m) SPFormer (79.07%).

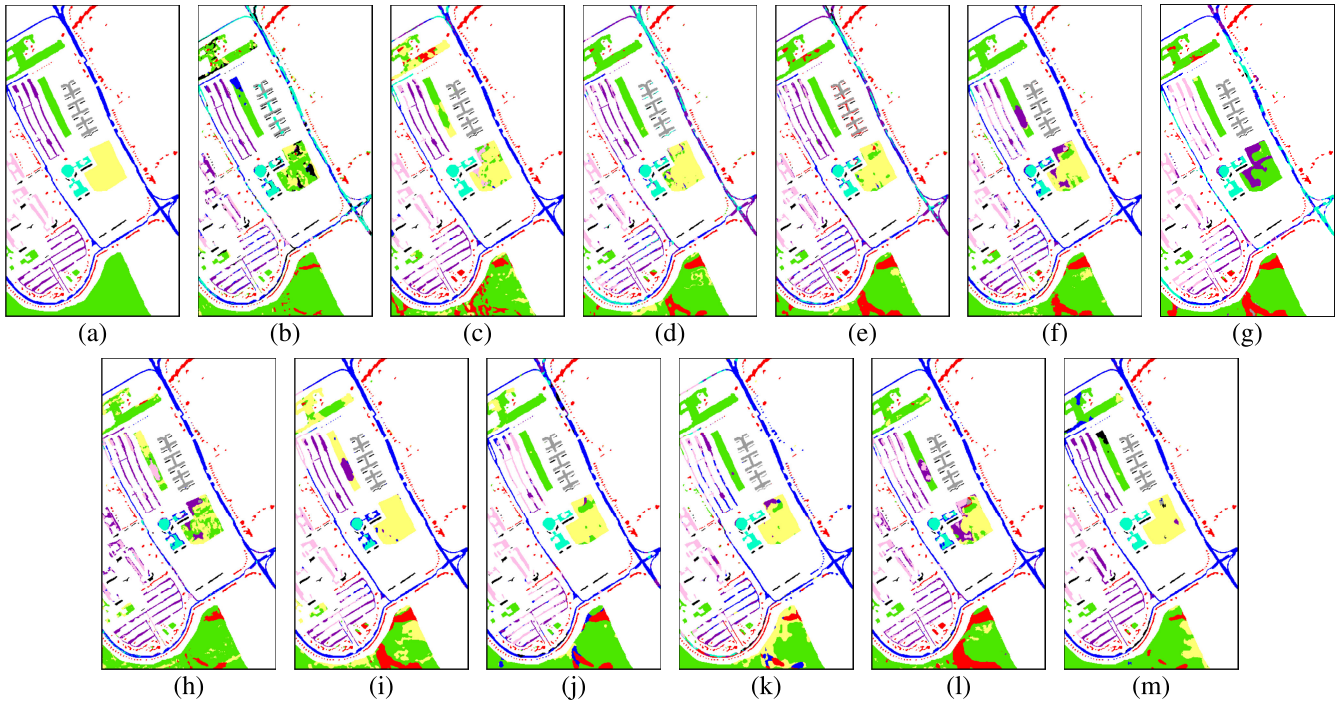


Fig. 11. Classification maps obtained by different methods in the University of Pavia (five labeled samples per class for training). (a) Ground Truth. (b) 3DCNN (64.76%). (c) SSRN (73.96%). (d) DFSL-NN (76.87%). (e) DFSL-SVM (77.41%). (f) DCFSL (81.86%). (g) DPN-HRA (70.27%). (h) RN-FSC (77.95%). (i) 3DAES (73.61%). (j) S3Net (84.73%). (k) SSFTT (78.53%). (l) GAHT (77.66%). (m) SPFormer (84.61%).

and University of Houston datasets, SPFormer has a greater advantage in classification accuracy than other models when dealing with limited training samples. While SPFormer does not achieve the highest classification accuracy among all methods for the University of Pavia dataset under the configuration of five training samples per class, it consistently outperforms other models as the number of training samples increases. Therefore, the above experimental results and analysis can powerfully prove the superiority of SPFormer in terms of generalization performance.

F. Classification Results

To verify the superiority of SPFormer in the context of HSI FSL, the classification accuracy results of different models under five training samples per class are reported in Tables V–VIII for Indian Pines, University of Pavia, Salinas, and University of Houston datasets, respectively. To reflect the classification performance of different models visually and intuitively, the ground truth and classification maps based on different models for four experimental datasets are displayed, respectively, in Figs. 10–13.

1) Indian Pines: The classification results of the Indian Pines dataset are presented in Table VIII. SPFormer achieves the best classification performance across three indicators, OA, AA, and $\kappa \times 100$, reaching values of 79.07 ± 3.39, 87.78 ± 1.90, and 76.38 ± 3.74, respectively. These results surpass other models by significant margins, with improvements ranging from 28.52% to 2.05% for OA, 25.12% to 1.53% for AA, and 31.78% to 2.19% for $\kappa \times 100$. In addition, SPFormer achieves the highest classification accuracy in six

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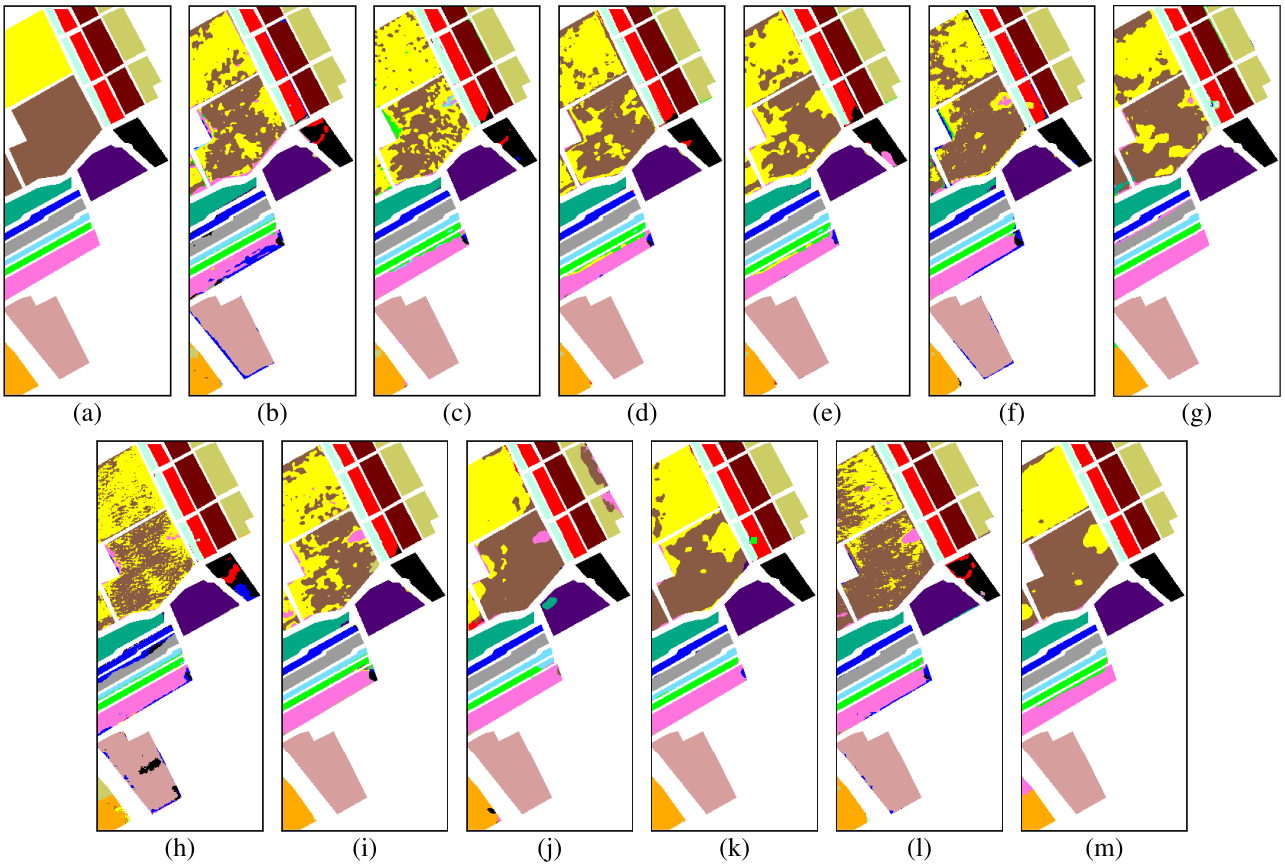


Fig. 12. Classification maps obtained by different methods in Salinas (five labeled samples per class for training). (a) Ground Truth. (b) 3DCNN (83.27%). (c) SSRN (86.80%). (d) DFSL-NN (88.61%). (e) DFSL-SVM (86.77%). (f) DCFSL (89.46%). (g) DPN-HRA (92.69%). (h) RN-FSC (83.34%). (i) 3DAES (87.17%). (j) S3Net (92.89%). (k) SSFTT (92.15%). (l) GAHT (87.15%). (m) SPFormer (93.74%).

of 16 land cover categories, especially for the Soybeans-clean category, in which SPFormer has achieved a more significant accuracy improvement compared to other models. The corresponding classification maps produced by different models are shown in Fig. 10, and the classification map generated by SPFormer exhibits better homogeneity within regions representing the same land cover categories. Especially for the Soybeans-mintill and Woods categories, which constitute a substantial portion of all samples, SPFormer exhibits significantly fewer fragmented areas compared to other models inside the image block.

2) University of Pavia: The classification results for the University of Pavia dataset are shown in Table IX. While SPFormer does not achieve the highest OA compared to the best-performing model, S3Net, the gap between them is marginal, only 0.12%. However, SPFormer outperforms other models in terms of AA and $\kappa \times 100$. We consider that the University of Pavia dataset has higher spatial resolution than the other two datasets with fewer bands, and it is easier to achieve higher classification accuracy for S3Net based on contrastive learning and CNN architecture, which can distinguish subtle differences between different land cover categories and extract richer local texture information of HSI blocks than the Transformer architecture model. The classification maps are shown in Fig. 11. The classification map generated by SPFormer exhibits superior homogeneity compared to other models in classifying the Bare soil category, which is represented by light yellow. In addition, SPFormer achieves higher classification accuracy in three of the nine categories present in the University of Pavia dataset.

3) Salinas: The classification results of the Salinas dataset are reported in Table X, and SPFormer achieves the highest accuracy across all three indicators: OA, AA, and $\kappa \times 100$. Specifically, SPFormer achieves a significantly higher classification accuracy for OA compared to other models, with improvement ranging from 10.47% to 0.85%. Besides, for AA and $\kappa \times 100$, SPFormer outperforms other models by 8.53% to 1.18% and 11.59% to 0.95%, respectively. What is more, SPFormer achieves the highest category classification accuracy in eight of 16 categories. As for the classification maps, which are shown in Fig. 12, SPFormer achieves smoother classification results. Especially for Grapes-untrained and Vinyard-untrained, which account for a substantial proportion of the total samples, the classification map generated by SPFormer demonstrates superior homogeneity in the corresponding regions representing these two land cover categories.

4) University of Houston: The classification results of the University of Houston dataset are reported in Table XI. Compared with other models, OA, AA, and kappa obtained by SPFormer have achieved at least 1.01%, 0.51%, and 1.11% accuracy improvements, respectively. In addition, SPFormer can achieve the highest classification accuracy on more land cover categories.

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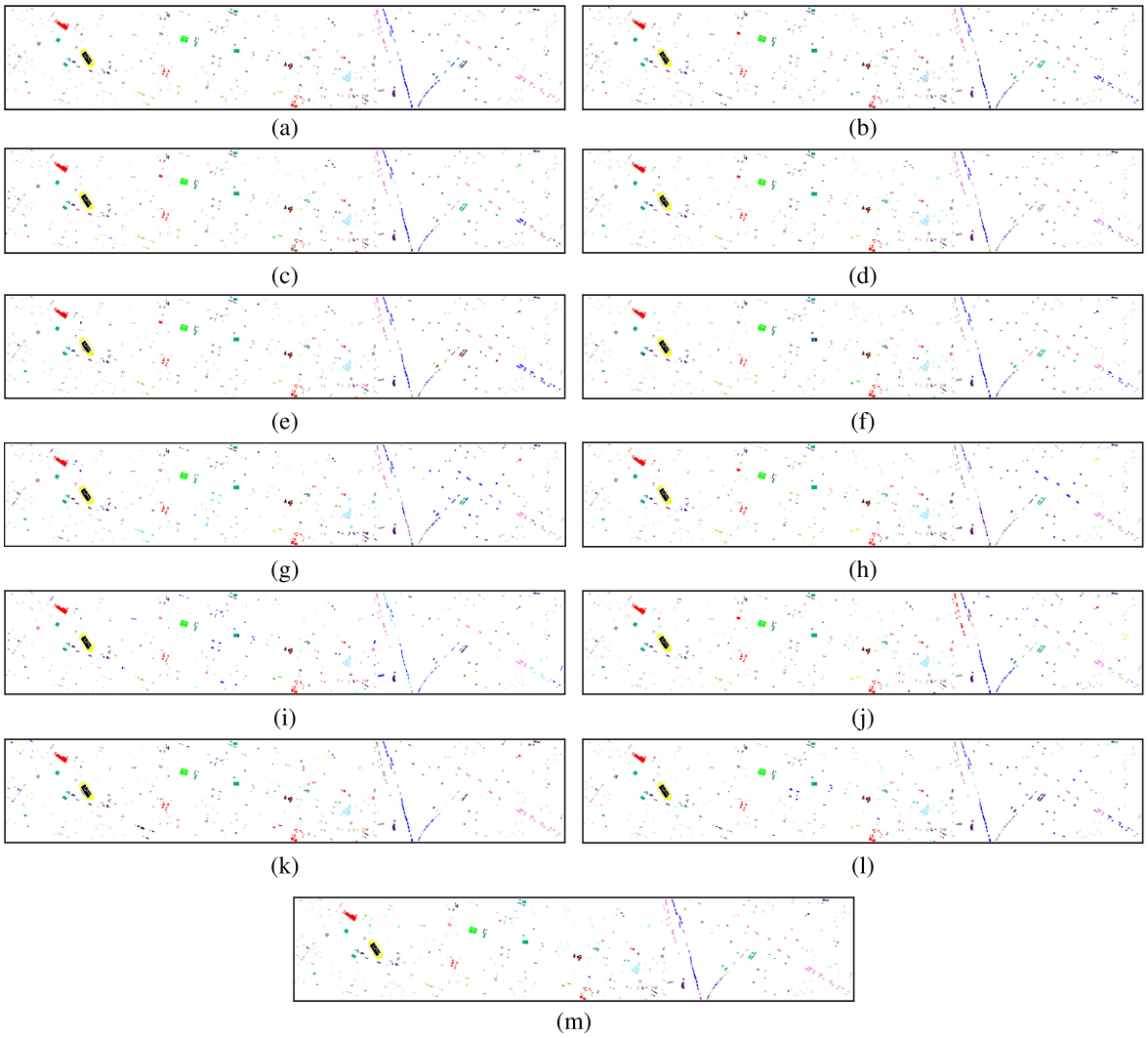


Fig. 13. Classification maps obtained by different methods in Houston (five labeled samples per class for training). (a) Ground Truth. (b) 3DCNN (67.13%). (c) SSRN (75.99%). (d) DFSL-NN (76.24%). (e) DFSL-SVM (76.44%). (f) DCFSL (75.91%). (g) DPN-HRA (67.09%). (h) RN-FSC (74.33%). (i) 3DAES (73.04%). (j) S3Net (73.93%). (k) SSFTT (75.54%). (l) GAHT (78.43%). (m) SPFormer (79.44%).

TABLE IX

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN UNIVERSITY OF PAVIA (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN [58]	SSRN [17]	DFSL-NN [59]	DFSL-SVM [59]	DCFSL [38]	DPN-HRA [46]	RN-FSC [60]	3DAES [54]	S3Net [55]	SSFTT [33]	GAHT [34]	**SPFormer (Ours)**
1	55.01±11.49	85.71±5.50	58.54±12.08	63.47±14.83	77.89±12.85	**91.84±5.35**	72.78±12.97	69.02±13.08	85.99±7.48	63.68±5.85	84.16±8.79	89.69±5.57
2	68.25±9.77	61.21±17.80	77.63±9.63	78.17±9.96	87.38±3.86	**94.54±4.09**	83.44±4.30	81.55±7.37	81.82±14.35	81.52±9.57	73.16±13.48	75.70±10.46
3	61.47±8.17	65.72±14.23	78.52±10.30	81.54±10.28	63.53±12.97	44.87±12.66	67.36±11.15	54.97±11.66	82.82±7.02	78.87±8.68	64.47±9.24	**83.21±9.43**
4	77.45±5.35	91.22±5.27	90.94±4.04	91.08±5.03	92.50±4.36	63.97±16.20	91.00±3.89	90.18±5.67	89.41±4.48	84.05±6.92	**93.98±5.31**	91.27±4.96
5	83.71±10.47	96.87±5.46	99.58±0.33	95.82±5.78	99.31±0.98	98.29±1.90	99.44±0.53	98.10±2.47	99.27±1.07	99.70±0.45	**99.94±0.16**	99.93±0.20
6	48.58±5.71	77.10±13.20	81.87±11.45	77.59±15.16	74.07±5.47	65.75±21.53	57.86±5.41	48.90±6.44	84.79±10.31	80.27±12.91	70.54±13.10	**93.63±8.76**
7	78.40±6.99	93.21±6.04	91.52±3.02	87.48±4.38	78.94±7.90	42.61±23.60	68.94±8.79	70.83±18.92	**99.77±0.40**	99.34±0.63	84.39±7.79	99.65±0.44
8	60.33±11.98	81.56±8.79	70.41±13.15	70.64±17.68	63.53±11.79	58.27±12.19	72.20±10.04	59.82±7.44	84.29±18.06	65.09±8.37	76.23±11.39	**89.78±6.65**
9	87.77±12.16	99.71±0.60	87.2±5.09	91.94±5.64	99.11±0.76	79.93±15.11	98.45±1.64	98.86±0.77	82.24±10.76	88.83±9.30	**99.47±0.60**	95.61±5.89
OA (%)	64.76±3.95	73.96±6.82	76.87±3.48	77.41±5.92	81.86±1.91	70.27±5.23	77.95±2.05	73.61±2.61	**84.73±5.57**	78.53±3.26	77.66±4.11	84.61±4.36
AA (%)	69.00±2.66	83.59±2.82	81.80±2.05	81.97±3.24	81.81±1.47	71.12±3.09	79.05±1.00	74.69±3.66	87.82±2.95	82.37±2.77	82.93±1.32	**90.94±1.66**
$\kappa\times100$	55.50±4.17	67.89±7.31	70.70±3.86	71.28±7.07	76.41±2.31	62.82±5.90	71.40±2.38	65.87±3.40	80.56±6.45	72.46±3.84	71.74±4.38	**80.67±5.07**

IV. DISCUSSION

In this section, complexity analysis between different models, as well as feature separability results for different variants of SPFormer, is discussed. In addition, some considerations about Transformer architecture models for hyperspectral FSL are given.

A. Complexity Analysis

The training time, the test time, and the model parameter number of different models with five samples per class on different datasets are reported in Table XII. Furthermore, if the model is not trained with an end-to-end strategy, the additional training time will also be reported in the corresponding model record, where the former time is for training autoencoder and the latter is for training backbone. The backbone of SPFormer exhibits moderate training and test time among all experimental models. Considering that both the transfer learning-based and the relation network-based methods need source domain data for pretraining an initial model, we think that SPFormer requires less training time than the former and less test time than the latter. Although SPFormer spends more time on training and testing than SSFTT and GAHT, the model parameter of SPFormer is only 20%–40% of SSFTT and 4%–8% of GAHT.

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TABLE X

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN SALINAS (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN [58]	SSRN [17]	DFSL-NN [59]	DFSL-SVM [59]	DCFSL [38]	DPN-HRA [46]	RN-FSC [60]	3DAES [54]	S3Net [55]	SSFTT [33]	GAHT [34]	**SPFormer (Ours)**
1	94.85±6.05	99.57±0.70	99.80±0.26	98.09±3.02	98.99±1.12	99.52±0.87	96.26±2.47	99.39±1.20	98.40±2.98	98.62±2.42	**99.85±0.37**	99.58±1.21
2	95.17±7.18	99.61±0.49	98.95±1.29	95.96±4.97	99.59±0.92	98.59±1.30	98.71±1.09	88.93±19.80	97.13±5.71	99.23±2.10	97.95±2.28	**100.00±0.00**
3	89.56±5.79	89.10±6.81	96.13±4.18	92.77±6.01	90.44±10.53	96.11±5.14	82.51±5.25	**98.44±2.36**	97.98±5.08	98.19±4.94	85.26±8.05	94.52±7.73
4	98.91±0.66	99.45±1.08	98.96±0.61	98.25±2.22	99.48±0.62	89.54±6.68	98.77±1.28	99.35±0.38	98.91±2.08	98.45±1.96	99.29±1.15	**99.94±0.08**
5	93.98±2.92	91.11±10.80	95.67±2.86	92.58±5.21	91.08±3.03	95.81±3.16	90.63±6.78	95.59±2.56	97.36±2.74	93.18±5.57	91.55±3.87	**97.65±1.86**
6	98.58±1.89	99.57±0.68	99.01±0.65	97.14±2.41	98.44±2.20	99.32±0.68	98.73±0.97	99.79±0.22	99.10±1.05	98.60±2.00	**99.99±0.02**	99.48±0.79
7	96.72±2.93	99.80±0.25	99.13±0.73	96.16±8.20	98.98±1.77	98.41±2.57	93.85±4.49	99.61±0.45	99.92±0.20	99.41±0.81	99.71±0.36	**99.94±0.07**
8	60.13±11.11	66.02±15.88	71.06±9.25	64.52±9.50	77.86±9.76	**92.73±4.12**	68.69±9.68	67.19±15.10	80.37±9.15	80.73±5.80	65.99±11.54	83.03±6.67
9	94.49±2.67	96.92±3.07	99.20±0.58	98.55±1.81	99.07±1.26	99.00±0.92	91.43±1.46	97.10±2.42	99.45±1.19	99.86±0.27	98.09±2.25	**100.00±0.00**
10	76.22±11.69	84.76±6.13	88.55±4.77	87.97±5.77	87.02±3.49	**94.39±4.45**	71.12±6.56	90.90±2.95	91.35±5.73	92.26±4.24	86.48±5.80	93.71±4.83
11	87.63±7.32	96.99±5.52	99.01±0.43	98.26±1.28	98.57±1.36	96.47±5.96	82.81±15.07	99.16±0.63	99.32±1.04	99.79±0.21	97.41±2.82	**99.94±0.12**
12	97.64±3.42	98.55±2.55	99.88±0.27	**99.91±0.08**	99.61±0.56	91.79±6.14	93.13±3.48	99.15±1.59	91.10±11.22	97.64±3.30	99.16±1.16	97.19±3.30
13	98.12±2.25	99.20±1.22	**100.00±0.00**	99.96±0.13	99.25±0.34	86.33±18.63	96.02±2.72	99.66±0.35	97.52±3.40	97.17±4.00	98.50±2.95	98.10±1.34
14	96.94±2.46	97.90±1.68	99.29±0.74	98.80±1.10	98.55±1.73	88.56±16.30	97.91±3.27	97.75±1.61	94.74±6.07	89.91±10.54	97.89±1.12	**99.85±0.21**
15	72.09±7.97	75.89±7.83	71.65±10.68	75.39±8.70	73.04±6.60	78.56±6.25	68.20±6.21	72.22±20.74	**90.65±14.73**	84.00±6.97	77.21±9.93	86.92±9.19
16	81.18±12.00	85.02±7.20	95.92±4.10	97.48±4.27	91.71±6.05	96.68±7.07	82.64±10.00	96.63±3.02	95.70±4.61	94.79±4.08	88.91±7.16	**97.95±2.48**
OA (%)	83.27±2.32	86.80±2.82	88.61±1.10	86.77±2.46	89.46±1.68	92.69±1.10	83.34±1.75	87.17±1.67	92.89±1.19	92.15±1.11	87.15±1.24	**93.74±1.18**
AA (%)	89.51±1.57	92.47±1.54	94.51±0.61	93.24±1.89	93.86±0.76	93.86±2.26	88.21±1.49	93.80±1.44	95.56±0.91	95.11±1.17	92.70±0.65	**96.74±0.75**
$\kappa\times100$	81.46±2.54	85.37±3.08	87.34±1.21	85.34±2.70	88.29±1.84	91.88±1.21	81.50±1.93	85.76±1.85	92.10±1.33	91.28±1.22	85.76±1.34	**93.05±1.31**

TABLE XI

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN UNIVERSITY OF HOUSTON (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN [58]	SSRN [17]	DFSL-NN [59]	DFSL-SVM [59]	DCFSL [38]	DPN-HRA [46]	RN-FSC [60]	3DAES [54]	S3Net [55]	SSFTT [33]	GAHT [34]	**SPFormer (Ours)**
1	87.78±5.36	86.57±9.17	93.56±5.79	**94.31±6.09**	88.92±6.40	86.52±7.70	85.69±7.33	91.27±6.42	74.40±11.38	73.73±12.00	90.26±6.51	85.27±10.59
2	74.96±7.59	80.94±7.93	**86.90±7.25**	86.82±6.97	79.03±11.23	77.25±15.13	84.88±6.66	73.61±15.97	86.58±6.10	85.36±7.90	83.51±9.90	82.02±9.35
3	78.12±37.26	**99.62±0.95**	94.83±5.10	92.90±6.06	96.53±3.19	98.67±2.24	86.01±7.12	96.79±3.19	99.35±0.98	96.49±2.45	94.61±8.24	99.58±0.62
4	88.32±3.95	92.08±3.53	89.01±3.03	89.53±2.77	90.79±3.21	81.81±13.16	**94.96±2.28**	88.52±6.31	91.34±5.78	82.21±10.02	94.37±2.22	92.72±2.89
5	93.19±5.05	93.51±9.01	98.63±2.88	98.80±2.91	95.24±3.80	89.28±8.74	90.49±11.72	86.24±9.13	88.24±7.92	94.02±7.78	98.96±1.65	**99.47±0.73**
6	74.41±5.00	87.91±7.00	78.66±4.39	82.47±5.09	80.97±6.58	**94.45±5.35**	77.94±5.67	84.09±4.57	90.00±5.80	85.97±7.41	87.62±5.63	87.69±4.31
7	58.99±23.23	75.35±9.10	66.38±7.06	74.24±8.34	63.06±13.90	63.49±17.86	72.38±9.24	63.46±10.45	63.61±8.70	64.97±10.48	76.96±5.25	**82.49±12.36**
8	35.59±9.80	34.88±9.66	42.40±7.21	47.91±11.93	44.67±5.01	**66.60±20.55**	39.59±19.77	58.10±7.29	41.98±7.86	42.82±4.57	47.94±9.50	38.64±8.23
9	62.32±8.95	73.24±10.47	55.12±8.46	65.82±11.66	63.18±9.36	44.30±13.42	59.05±9.02	64.02±8.42	51.11±4.47	55.17±16.35	**79.00±3.36**	78.60±7.99
10	38.89±15.86	61.31±17.88	64.16±16.79	53.58±15.06	66.04±14.39	47.61±10.97	63.86±17.38	54.49±10.05	64.79±18.13	**75.14±14.98**	55.87±10.96	70.23±20.70
11	57.09±11.22	66.22±9.92	64.51±6.29	58.48±20.08	62.93±8.75	62.02±23.41	56.19±9.69	56.40±12.12	79.18±10.03	76.86±13.44	65.20±5.35	**79.59±7.31**
12	43.65±8.33	52.93±11.99	66.66±7.58	60.11±19.01	**73.66±11.54**	55.81±15.07	60.78±16.47	62.75±13.37	50.73±10.76	65.22±11.01	55.12±12.25	51.88±10.20
13	63.58±8.75	90.04±4.29	93.53±0.48	92.39±2.49	86.66±7.66	54.87±18.20	95.60±1.47	68.03±9.67	90.91±3.36	90.88±5.72	**95.73±0.50**	84.14±11.35
14	89.67±7.18	99.79±0.56	96.24±1.92	95.18±3.89	88.30±8.13	95.29±5.21	96.45±3.91	88.35±7.43	**100.00±0.00**	99.57±0.91	99.48±0.54	99.95±0.09
15	96.44±2.02	100.00±0.00	96.21±4.65	97.42±2.76	95.73±2.57	85.26±8.17	97.15±2.50	95.97±6.13	99.85±0.46	97.85±5.01	100.00±0.00	**100.00±0.00**
OA (%)	67.13±3.50	75.99±2.33	76.24±1.22	76.44±3.08	75.91±2.67	67.09±3.58	74.33±2.44	73.04±4.45	73.93±2.72	75.54±2.14	78.43±1.45	**79.44±2.13**
AA (%)	69.53±3.35	79.63±1.68	79.12±1.18	79.33±2.53	78.38±2.41	73.55±3.53	77.40±1.81	75.47±4.05	78.14±2.22	79.08±1.68	81.64±1.39	**82.15±1.99**
$\kappa\times100$	64.51±3.73	74.11±2.50	74.33±1.32	74.54±3.33	73.97±2.88	64.49±3.85	72.28±2.63	70.89±5.10	71.87±2.93	73.58±2.31	76.70±1.57	**77.81±2.30**

TABLE XII

RUNNING TIME (S) AND PARAMETER SIZE OF DIFFERENT METHODS

Datasets	Metrics	3DCNN [58]	SSRN [17]	DFSL-NN [59]	DFSL-SVM [59]	DCFSL [38]	DPN-HRA [46]	RN-FSC [60]	3DAES [54]	S3Net [55]	SSFTT [33]	GAHT [34]	**SPFormer (Ours)**
Indian Pines	Training time (s)	2.12	8.19	435.05	435.05	3821.83	421.63	485.29	201.78+58.32†	5.84	9.32	10.83	7.37+11.55†
	Testing time (s)	0.83	1.9	1.85	6.15	29.44	179.98	4.37	32.02	5.49	1.42	1.88	3.08
	Parameters	313964	540808	34920	34920	4268994	2638021	142177	269584	164464	218504	972624	86296
University of Pavia	Training time (s)	1.47	3.5	401.36	401.36	3568.57	78.72	483.72	2010.71+38.92†	5.23	4.64	7.01	320.52+6.12†
	Testing time (s)	2.9	4.74	6.74	5.92	31.93	277.09	13.83	69.76	20.23	5.38	8.47	8.07
	Parameters	89457	291801	34920	34920	4259294	1162213	142177	138128	100973	218049	927113	45225
Salinas	Training time (s)	2.39	8.52	483.58	483.58	3756.48	456.19	486.98	629.63+45.02†	16.91	11.30	12.18	40.72+9.76†
	Testing time (s)	4.72	10.36	9.04	16.31	33.21	1126.25	25.54	81.34	28.44	10.08	11.25	16.43
	Parameters	320364	551176	34920	34920	4269394	2638021	142177	269584	164464	218504	972624	86456
University of Houston	Training time (s)	2.36	20.14	427.33	427.33	3683.13	421.22	484.34	3996.67+49.51†	15.33	5.89	7.35	564.77+13.15†
	Testing time (s)	2.43	2.23	2.87	5.84	30.13	321.74	5.59	34.48	8.17	0.69	2.62	3.53
	Parameters	210363	395631	34920	34920	4264360	2637957	142177	751823	117851	153224	973583	46735

Note: † denotes the time for training autoencoder (former) and backbone (latter).

Several reasons are given to explain the superiority of SPFormer in terms of training time, test time, and model parameters. First, SPFormer employs a nonconvolutional model architecture, which is beneficial as the input–output data operations associated with convolution tend to be time-consuming. Second, the core module of SPFormer, CSSM-MHSP, is implemented by matrix multiplication, which is run efficiently under GPU acceleration and does not add additional parameters to the model. Finally, Transformer can be regarded as an attention-based model. SPFormer, on the other hand, employs a parameterless token mixer module based on cosine similarity to implement an attention mechanism, in contrast to the conventional scaled dot-product attention that necessitates learning a substantial number of parameters.

Overall, SPFormer achieves great performance for few-shot HSI classification without consuming excessive computing resources and occupying excessive storage resources for training.

B. Feature Separability

To further verify the effectiveness of the core module CSSM-MHSP and CTM in SPFormer, we conduct experiments on feature separability based on the results of ablation study 2. This study involves different model variants: the baseline variant without using CSSM-MHSP and CTM modules, variants using only CSSM-MHSP or the CTM module, and the full SPFormer variant incorporating both CSSM-MHSP and CTM modules. We extract features of HSI patches from the output of the GAP layer in each model variant. The dimensionality of the extracted feature is set to 40 for the Indian Pines and Salinas datasets and 32 for the University of Pavia and University of Houston datasets. Then, feature separability results acquired from different model variants are analyzed from qualitative and quantitative perspectives.

1) For Qualitative Analysis: We use PCA to map the output features into a 2-D plane and mark samples of different classes with points of different colors in the feature plane. The feature separability maps acquired by different variants of the proposed model are shown in Fig. 14, where (a)–(d), (e)–(h), (i)–(l), and (m)–(p) represent the results acquired from Indian Pines, the University of Pavia, Salinas, and the University of Houston, respectively. Some details can be inferred from the feature separability maps to verify the effectiveness of SPFormer for few-shot HSI classification. For the Indian Pines dataset, gray samples are more clustered on the map produced

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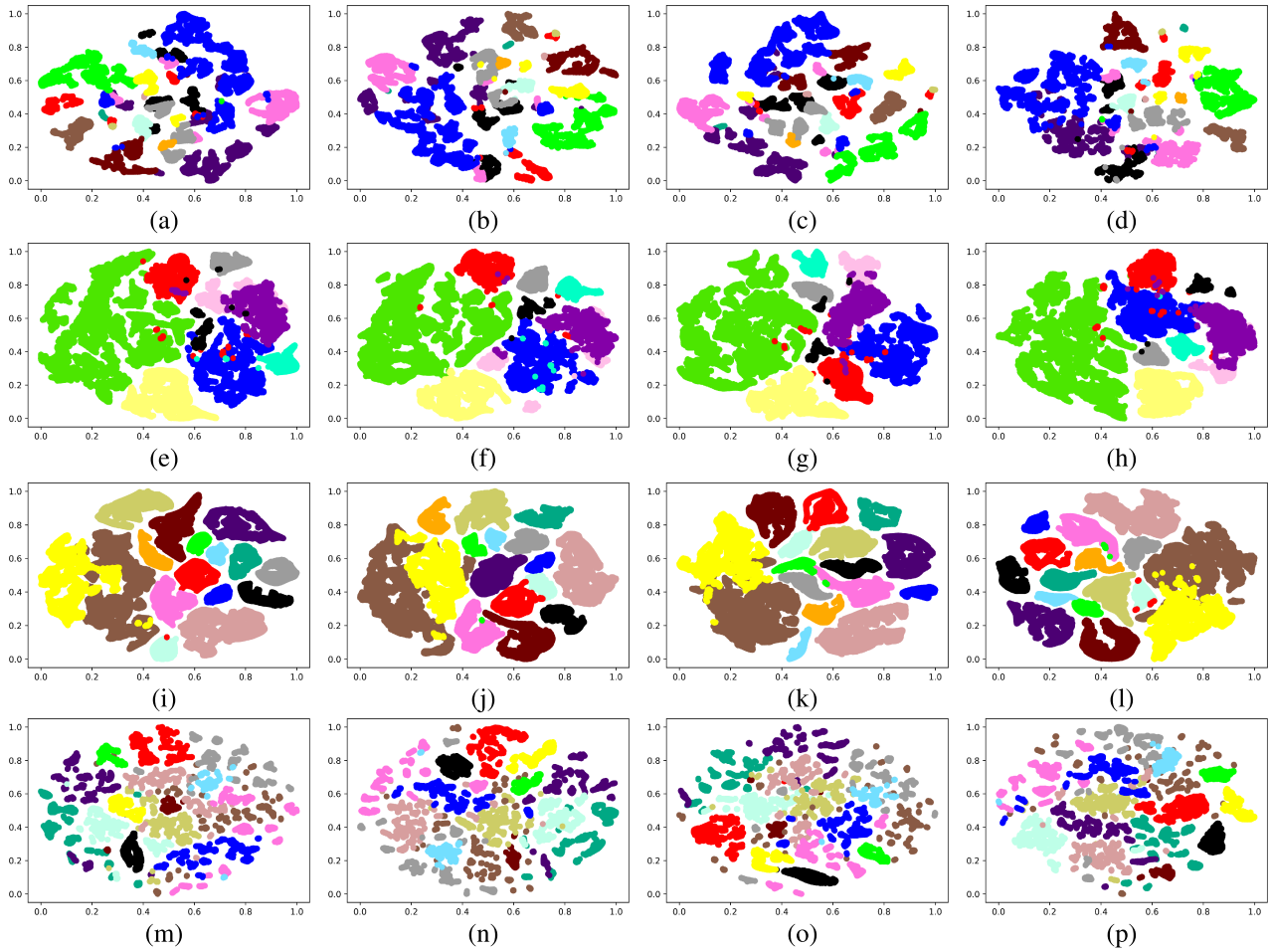


Fig. 14. Feature separability for different variants of SPFormer (five labeled samples per class for training). (a) Baseline ($S_b = 1.26$). (b) CTM ($S_b = 1.60$). (c) MHSP ($S_b = 2.34$). (d) SPFormer for Indian Pines ($S_b = 2.67$). (e) Baseline ($S_b = 1.56$). (f) CTM ($S_b = 1.61$). (g) MHSP ($S_b = 1.84$). (h) SPFormer for the University of Pavia ($S_b = 1.96$). (i) Baseline ($S_b = 1.67$). (j) CTM ($S_b = 1.69$). (k) MHSP ($S_b = 1.70$). (l) SPFormer for Salinas ($S_b = 1.75$). (m) Baseline ($S_b = 0.90$). (n) CTM ($S_b = 0.95$). (o) MHSP ($S_b = 1.04$). (p) SPFormer for the University of Houston ($S_b = 1.05$).

by SPFormer compared with other variants. For the University of Pavia dataset, the feature separability map acquired by SPFormer exhibits less intraclass differences for the black samples compared to the baseline and variants that only utilize the CSSM-MHSP module. Similarly, for the pink samples, SPFormer exhibits decreased intraclass differences compared to variants that exclusively employ the CTM module. For the Salinas dataset, SPFormer has greater interclass dispersion on feature maps than other model variants. For samples representing the Trees category in the University of Houston dataset, which is shown in cyan, SPFormer has a greater degree of intraclass aggregation than other variant models.

2) For Quantitative Analysis: We use the Fisher discriminant analysis method to calculate the interclass dispersion coefficient, which is used to compare the dispersion of features extracted by different model variants, and the corresponding calculation results are displayed below the feature separability map. The formula for the interclass dispersion coefficient is given as follows:

$$\frac{\sum_{i=1,j=1,i\neq j}^{c}\left(m_i - m_j\right)\left(m_i - m_j\right)^T}{c-1} \tag{17}$$

where m_i and m_j represent the mean features of the ith and jth classes, respectively, and c represents the number of categories in the corresponding dataset. The rules presented on the four datasets indicate that SPFormer performs better than only using the CSSM-MHSP module, followed by only using the CTM module and the baseline.

C. Innovative Contribution and Future Perspectives

Compared with CNN, the Transformer can sensitively capture global features of images, which is the most significant advantage of the Transformer architecture model. While the core of Transformer, MHSA and FFN modules, needs to learn a large amount of parameters, which is a training burden for few-shot HSI classification, the proposed SPFormer does not rigidly follow the original Transformer's module design method in the corresponding position of the core modules. Specifically, CSSM-MHSP and CTM modules rather than traditional MHSA are used, which are parameterless and have the property of central symmetric. In our work, the traditional antibottleneck structure of FFN in the Transformer is not used, and it is replaced by channel embedding composed of the same-dimension linear map and a GELU activation function, which is more lightweight. At the same time, considering that the number of network layers is shallow, the residual connection is not used in the above two modules.

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V. Conclusion

In this article, we propose a lightweight SPFormer for few-shot HSI classification. First, an autoencoder trained by self-supervised learning is used to reduce the dimensionality of HSI. Second, a parameter-free CSSM-MHSP module is designed to map spectral features to a more distinguishable feature space. Then, the CTM module is proposed to realize interactions between different tokens in the spatial dimension. Finally, GAP in conjunction with a fully connected layer is used for classification. SPFormer is a model specially designed for few-shot HSI classification. Different from most of the existing works in the field of few-shot HSI classification, SPFormer does not use any data augmentation trick. Besides, the commonly used contrastive learning or transfer learning paradigm is also applied in SPFormer. However, experimental results on four HSI datasets show that SPFormer can achieve competitive or even better classification performance compared with state-of-the-art methods under the condition of extremely limited training samples.

Although the proposed SPFormer has obtained great experimental results, we will further explore the CTM module to enhance the information interactions between tokens in the spatial domain.

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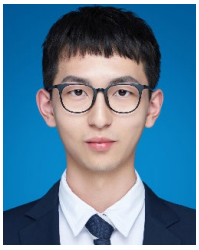


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